MILITARY STANDARD

MIL-STD 1325-157 A

RAILCAR LOADING OF HAZARDOUS MATERIALS

(NAVY)

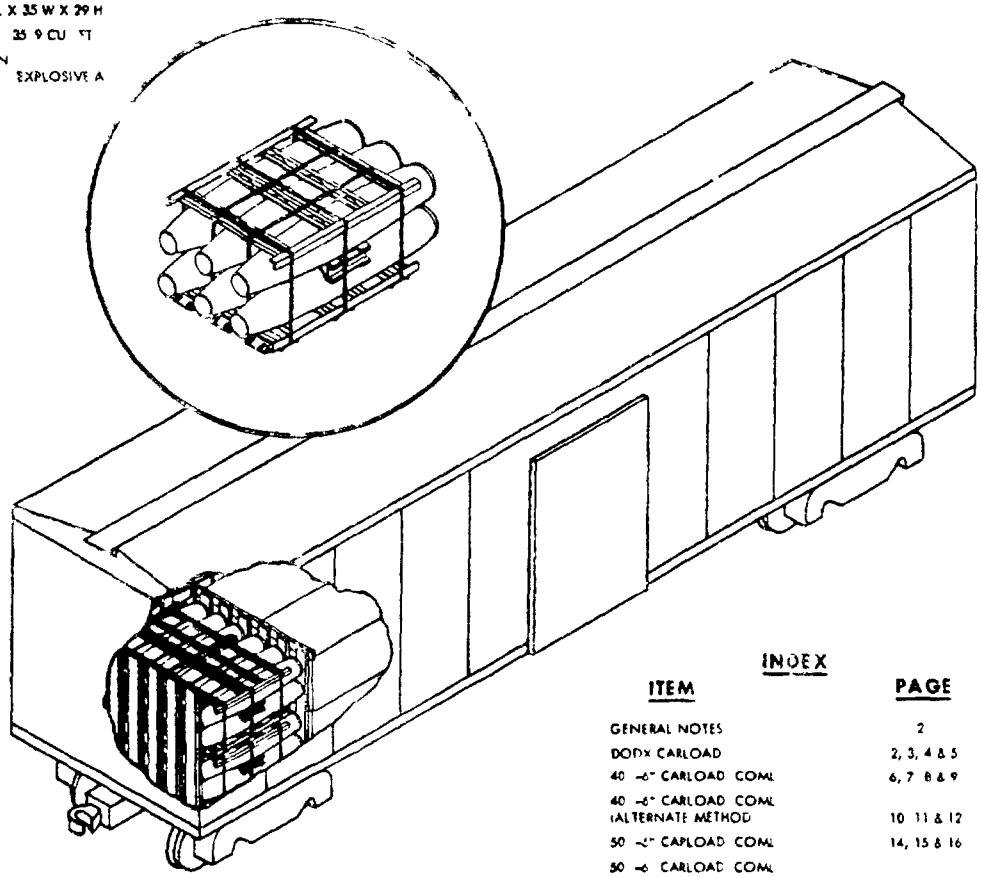
3 [illegible] MK 82 MOD 3 (THERMALLY PROTECTED)
ON PALLET, MHU-122/E
FLEET ISSUE UNIT LOAD

7 SEPTEMBER 1973

SUPERSEDING
MIL-STD-1325-157
9 JANUARY 1973

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/239
WEIGHT	3094 LBS (EST)
DIMENSIONS	61-3/8 L X 35 W X 29 H
CUBE	35.9 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVE A



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINER LOADING MILITARY STANDARDS, REFER TO DS MIL-HDBK-726 (NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3, 4 & 5
40 -6" CARLOAD COML	6, 7 8 & 9
40 -6" CARLOAD COML (ALTERNATE METHOD)	10 11 & 12
50 -6" CAPLOAD COML	14, 15 & 16
50 -6 CARLOAD COML (ALTERNATE METHOD)	18 THRU 23
50 -6" CARLOAD COML (ALTERNATE METHOD)	24 THRU 29
50 -6" CARLOAD COML [illegible] LOAD DIVIDERS	30 THRU 34
DUMMY UNIT LOAD	36 & 37
ALTERNATE SWAY BRACE FRAMES	38

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

[illegible] NWHL 12/14
DESIGNATED TECHNICAL DIRECTION AGENT (TDA) DATE

[illegible] 9/17/73
DESIGNATED SUPERVISORY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

[illegible]

DATE 8/14/73

ORIGINATOR [illegible] Aug 73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1 FOR GENERAL INFORMATION CONCERNING CLEANING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD 1325 (NAVY)

2 WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT UTILITY LOADER CARS SHALL BE SERIES DODX 53600

3 LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH

4 THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS

5 IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR

6 THE LOAD CONSISTS OF 500 LB MK 82 MOD 2 (THERMALLY PROTECTED) BOMBS PALLETIZED IN ACCORDANCE WITH WR-34/237

7 THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK

8 UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY)

9 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER - FED SPEC MM-L-751

NAILS - FED SPEC FF-N-105

STRAPPING - FED SPEC QQ-S-781, TYPE I, HEAVY DUTY CLASS A, DRY (UNLUBRICATED)

SEALS - FED SPEC OO-S-781, STYLE III, HEAVY DUTY

10 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1)

50 FT 6 IN. BOXCAR, DODX

1 THE CARLOAD CONSISTS OF 33 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING HOWEVER, THIS QUANTITY SHALL BE ADJUSTED BY SUBSTITUTING DUMMY UNIT LOADS (SEE PAGE 36) AS REQUIRED TO COMPLY WITH LOAD LIMIT OR WEIGHT DISTRIBUTION RULES OF THE AAR

2 IF UNABLE TO LOAD DOORWAY STACK AS SHOWN, SEE DETAIL G, PAGE 5 FOR ALTERNATE DOORWAY STACK LOCATION

3 A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750

4 TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING

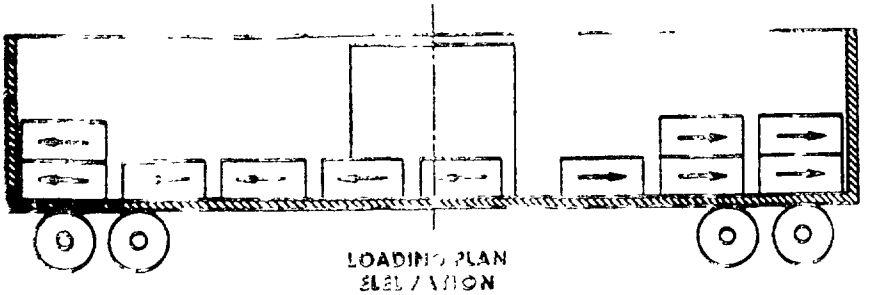
5 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE MIL-STD-1325 (NAVY)) EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2400 LBS OF THE LADING

* CLINCHED

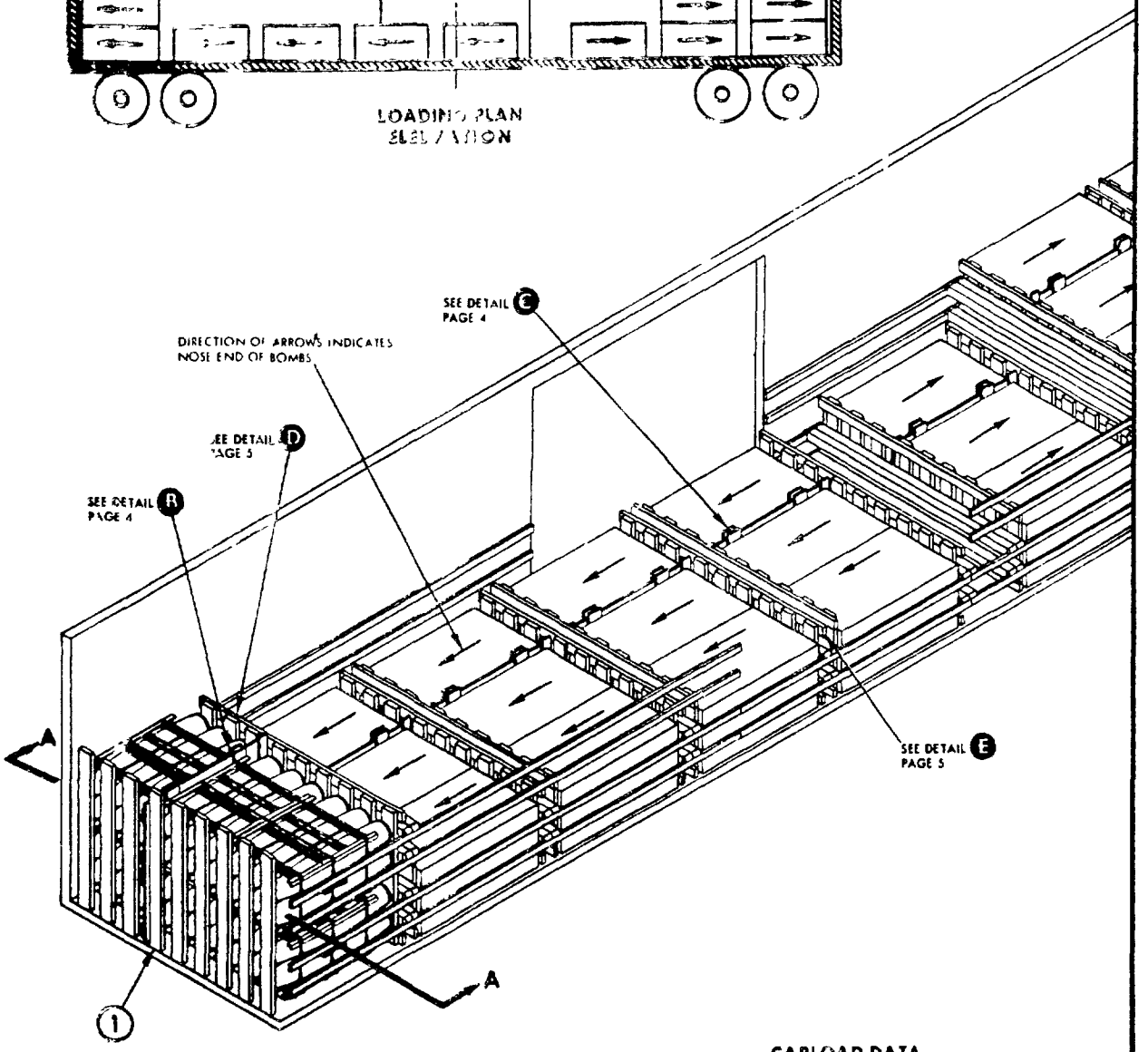
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NO NAILS REQD	NAIL
11	SWAY BRACE VERTICAL	1 x 6 x 34 2	20	10	3 PER JOINT	6d
10	SWAY BRACE HORIZONTAL	1 x 6 x 64	10	SEE 9	-	-
9	SWAY BRACE VERTICAL	1 x 4 x 34 1/2	10	10	3 PER JOINT	6d*
8	TIE PIECE	2 x 4 x 107	20	SEE 7	-	-
7	SEPARATOR GATE VERTICAL	2 x 6 x 36	90	8	3 PER JOINT	10d
6	TIE PIECE	2 x 4 x 107	8	SEE 5	-	-
5	SEPARATOR GATE VERTICAL	2 x 6 x 60	30	5	3 PER JOINT	10d
4	SWAY BRACE VERTICAL	1 x 6 x 63 1/2	12	3	3 PER JOINT	6d
3	SWAY BRACE HORIZONTAL	1 x 6 x 64	12	SEE 2	-	-
2	SWAY BRACE VERTICAL	1 x 4 x 63 1/2	6	3	3 PER JOINT	6d*
1	END WALL MEMBER	2 x 6 x 72	16	CAR WALL	1 PER JOINT	10d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



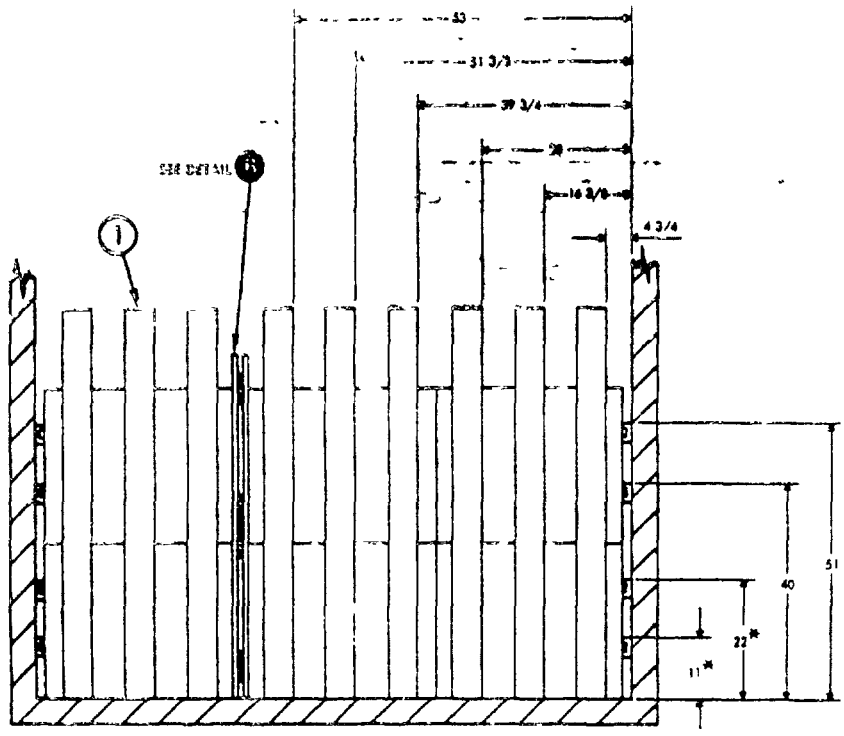
LOADING PLAN
ELEVATION



CARLOAD DATA

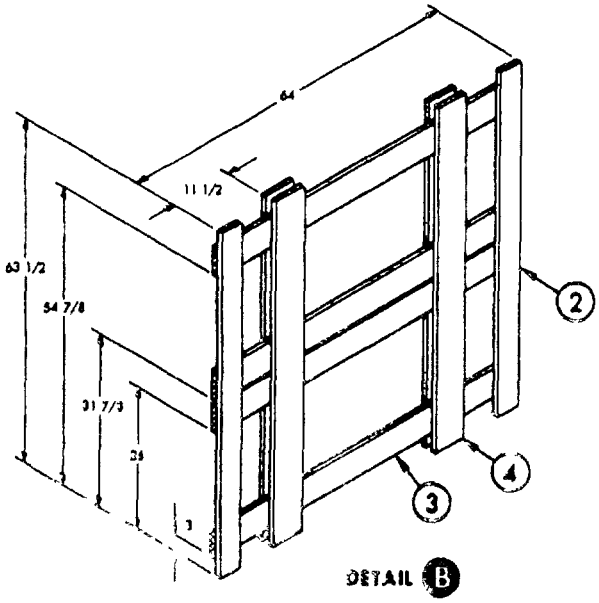
NUMBER OF UNIT LOADS	33
NUMBER OF CROSSMEMBERS REQ'D	44
LOAD WEIGHT	102,102 LBS (EST.)
DUNNAGE WEIGHT	2,159 LBS
CARLOAD WEIGHT	104,261 LBS (EST.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



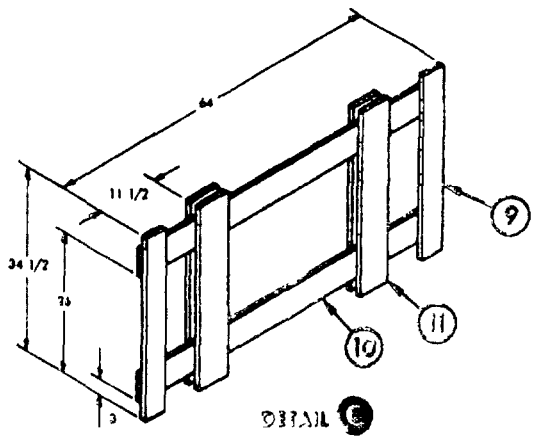
SECTION A-A

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS
ALSO LOCATION OF END WALL MEMBERS
*DOORWAY MEMBER LOCATIONS



DETAIL B

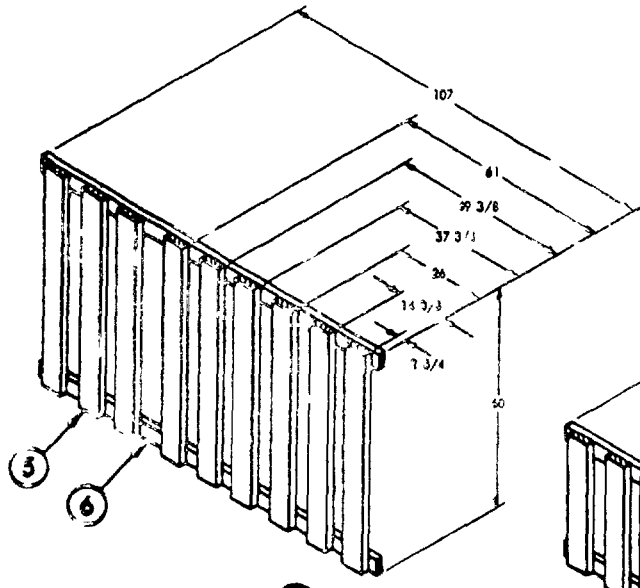
SWAY BRACE FRAME
3 REQUIRED



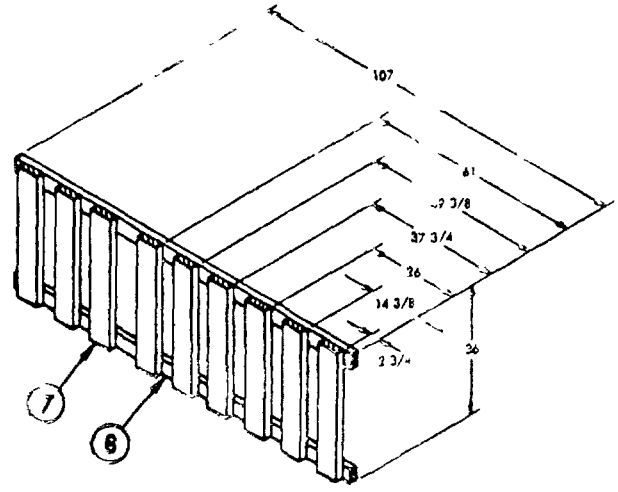
DETAIL C

SWAY BRACE FRAME
5 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

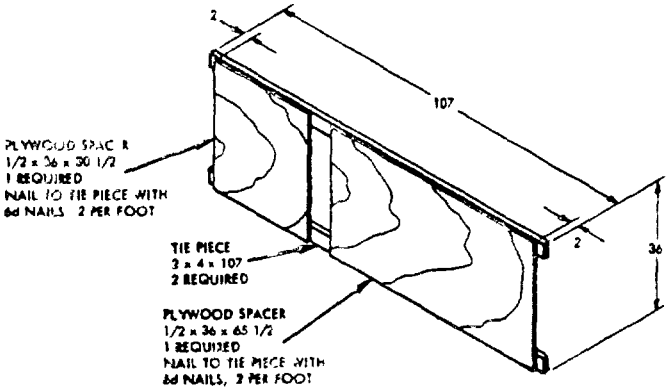


DETAIL D
SEPARATOR GATE
4 REQUIRED

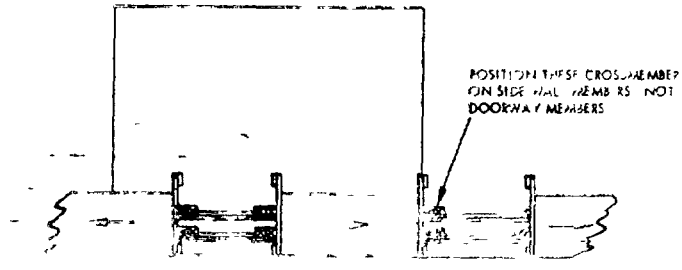


DETAIL F
SEPARATOR GATE
10 REQUIRED

(SEE ALTERNATE SEPARATOR GATE DETAIL F)



DETAIL F
ALTERNATE SEPARATOR GATE
10 REQUIRED



DETAIL G

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

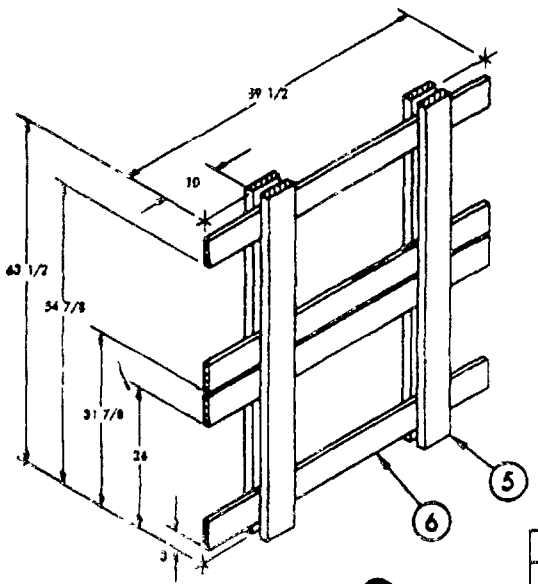
40 FT 6 IN. BOXCAR, COMMERCIAL

1 THE CARLOAD CONSISTS OF 36 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE. PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH MIL-STD-1365 (NAVY)

4 WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT 2 IN. USE ALTERNATE SWAY BRACE FRAMES SHOWN ON PAGE 38 IN PLACE OF DETAIL A



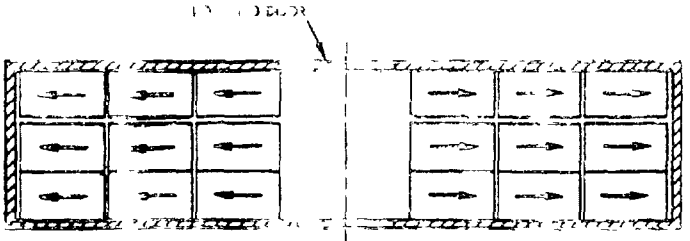
DETAIL A
SWAY BRACE FRAME
6 REQUIRED

* 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4 S

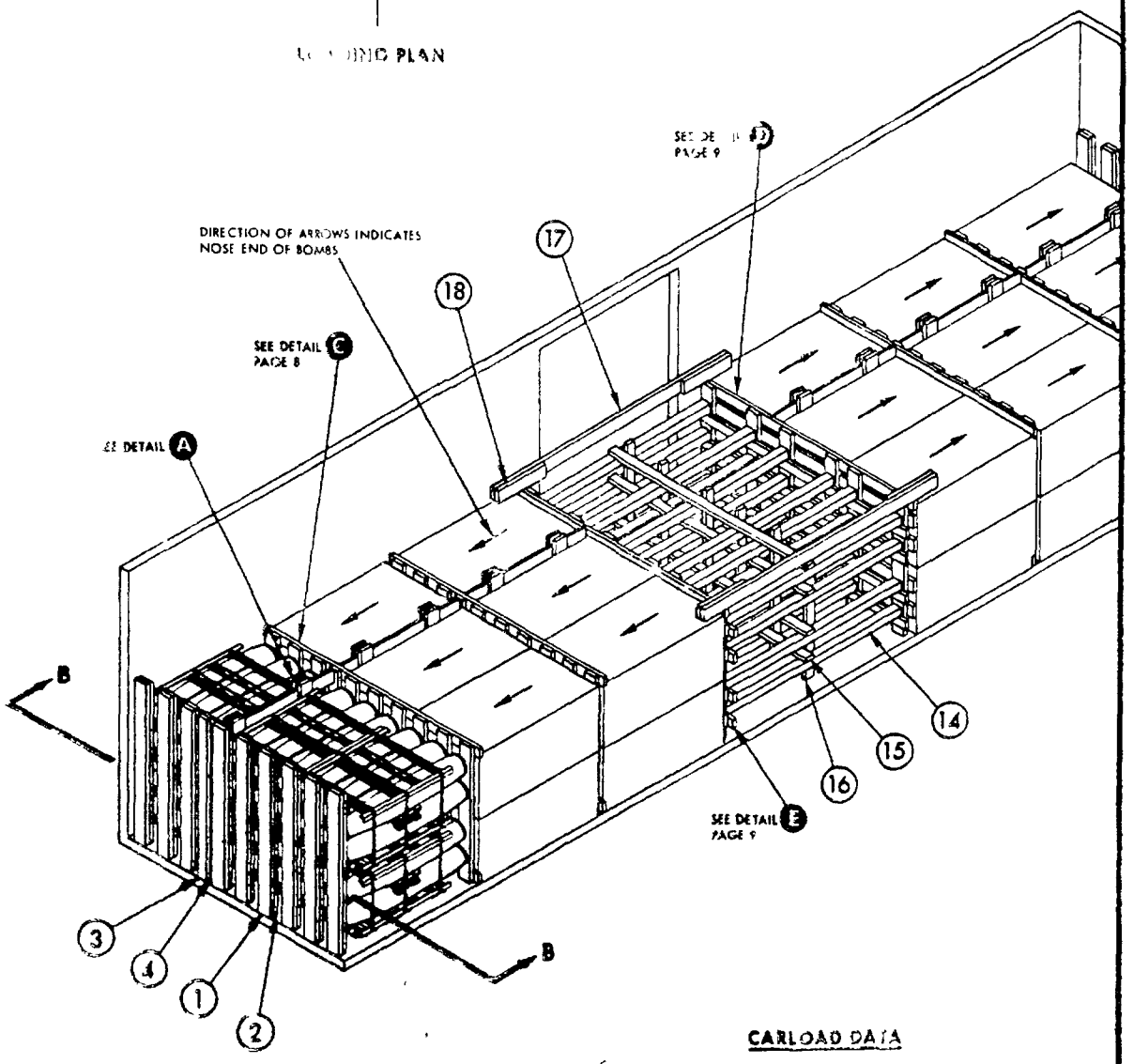
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
18	HOLD DOWN CLEAT	2 x 6 x 24	4	17	5	10d
17	GATE HOLD DOWN	2 x 6 x 10 PT	2	CAR WALL	5 EACH END	10d
16	VERTICAL TIE BAR	4 0	*	14	2 PER JOINT	16d
15	HORIZONTAL TIE BAR	* 4 CAR WIDTH-1	4	14	2 PER JOINT	16d
14	STRUT	4 x 4 x WEDGE FIT*	24	11	2 PER JOINT	16d
13	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 60	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 4 C	4	SEE *	-	-
8	SEPARATOR GATE HORIZONTAL	* x 4 CAR WIDTH-1	*	* 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	* * 40	36	SEE 8	-	-
6	SWAY BRACE HORIZONTAL	1 x 6 x 39 1/2	24	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 63 1/2	24	6	3 PER JOINT	10d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



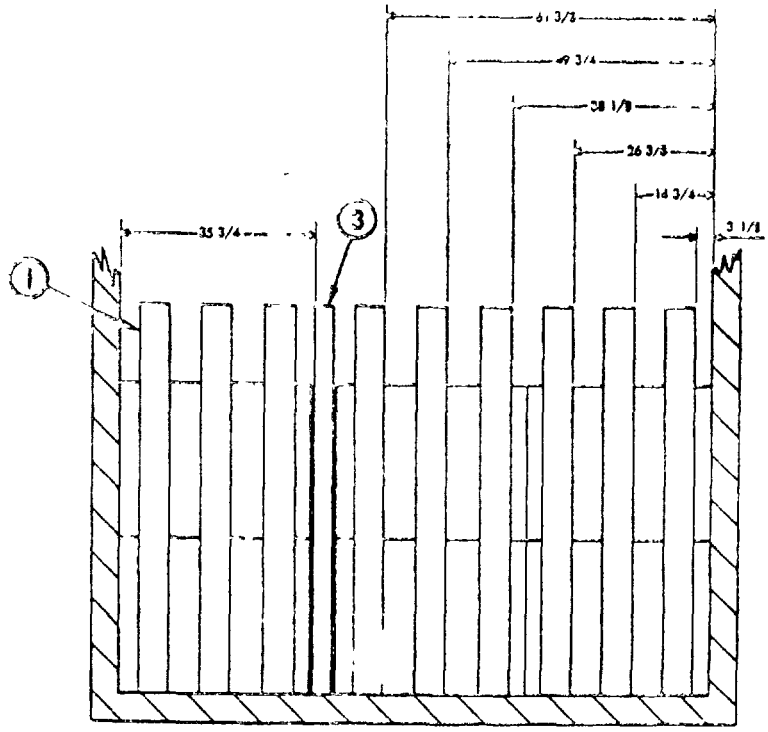
LOADING PLAN



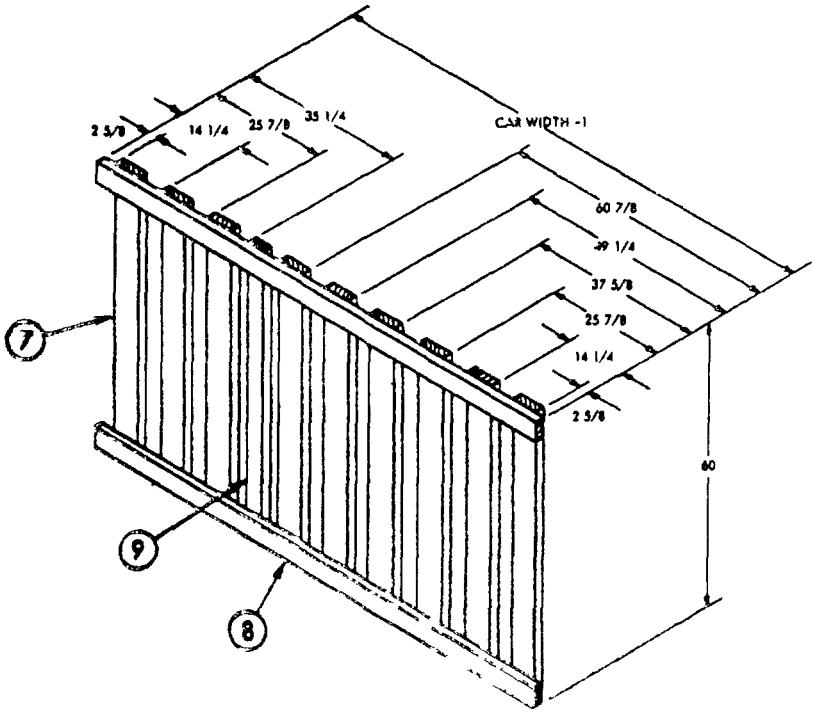
CARLOAD DATA

NUMBER OF UNIT LOADS	36
LOAD WEIGHT	111,284 LBS (NET)
DUNNAGE WEIGHT	2,945 LBS
CARLOAD WEIGHT	114,229 LBS (GROSS)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

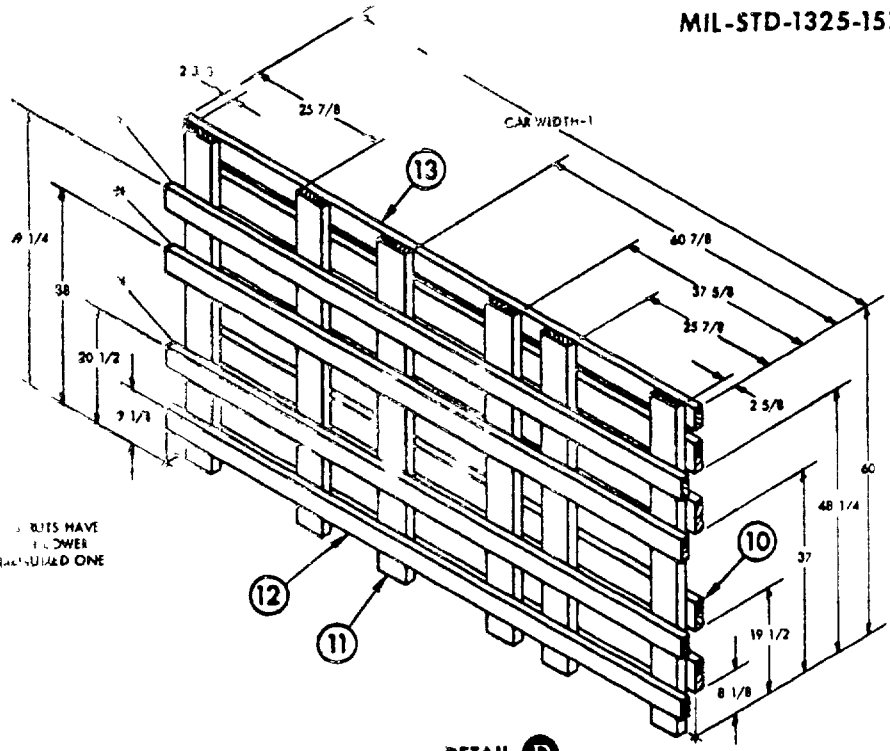


SECTION B-B
SHOWING LOCATION OF END WALL MEMBERS



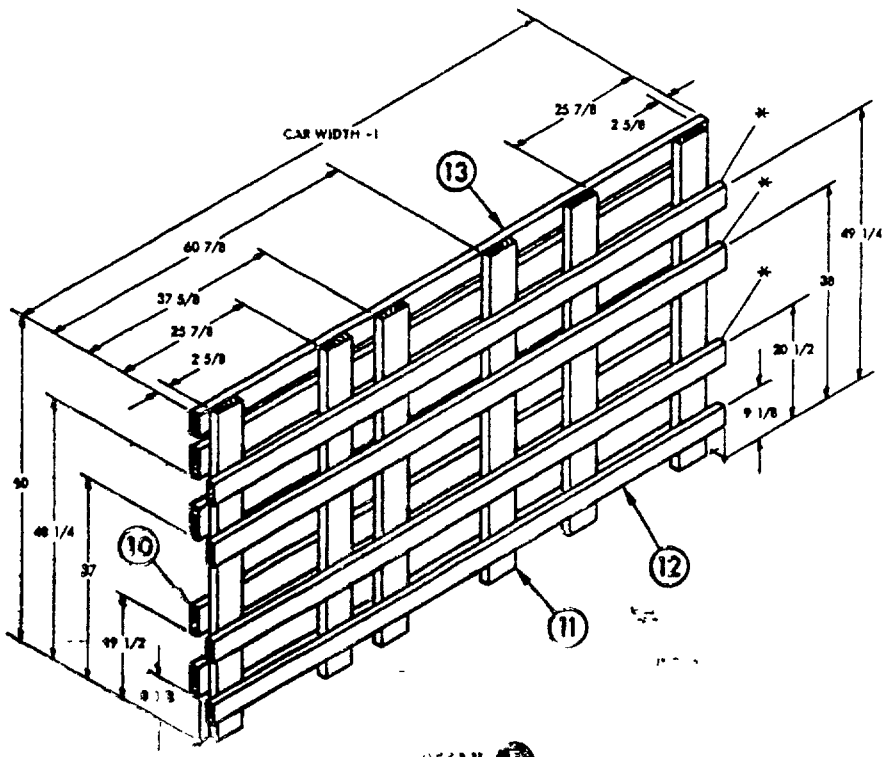
DETAIL 3
SEPARATOR PLATE
ASSEMBLED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



*POSITION AFT ... RUTS HAVE BEEN INSTALLED ... LOWER STRUT CLEAT ... REQUIRED ONE GATE ONLY)

DETAIL D
CENTER GATE
1 REQUIRED

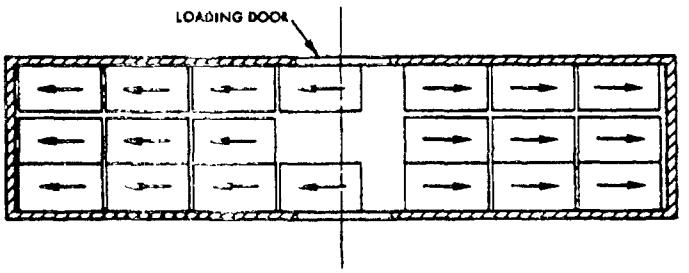


DETAIL E
CENTER GATE
1 REQUIRED

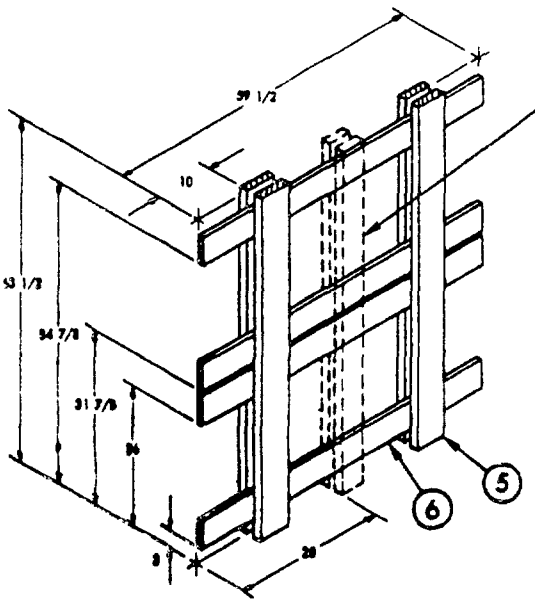
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

40 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.
4. WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT 2 IN USE ALTERNATE SWAY BRACE FRAMES SHOWN ON PAGE 38 IN PLACE OF DETAIL A.



LOADING PLAN



DETAIL A
SWAY BRACE FRAME
7 REQUIRED

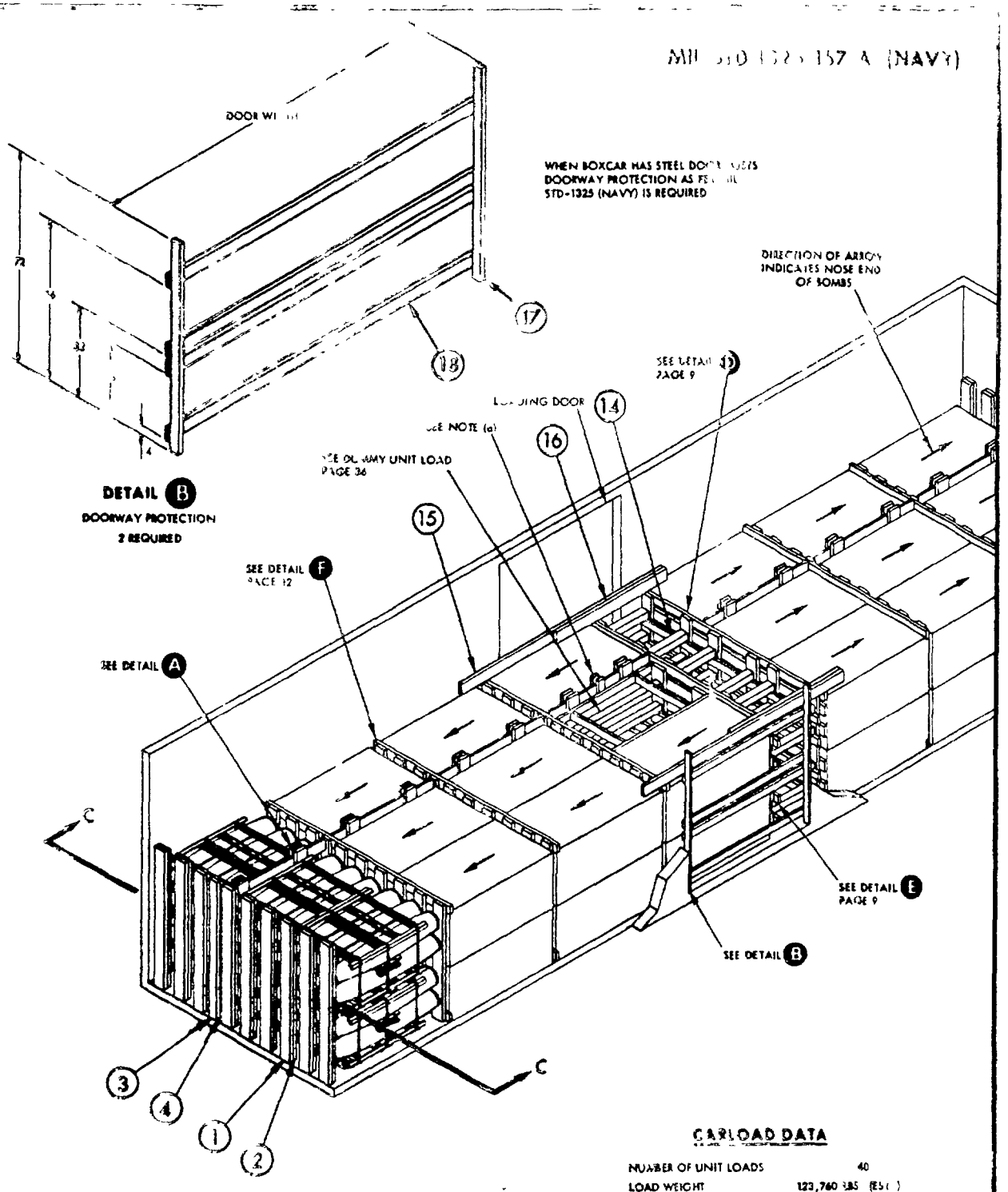
NOTE
(a) WHEN POSITIONING DUMMY LOAD NEXT TO SWAY BRACE FRAME 2 x 4 SPACERS MUST BE INSTALLED ON EACH SIDE OF FRAME. THIS IS NOT REQUIRED WHEN 3 DUMMY LOADS ARE LOADED ACROSS THE CAR.

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
18	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	17	2 PER JOINT	10d
17	VERTICAL DOORWAY MEMBER	2 x 3 x 72	4	DOOR POST	2 PER FOOT	20d
16	HOLD DOWN CLEAT	2 x 6 x 66	2	15	2 PER FOOT	10d
15	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
14	STRUT	4 x 4 x WEDGE FIT*	24	11	2 PER JOINT	16d
13	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	11	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 60	12	SEE 10		-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 60	5	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	10	7, 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 60	45	SEE 8	-	-
6	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/2	28	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 63 1/2	28	6	3 PER JOINT	10d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	7	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

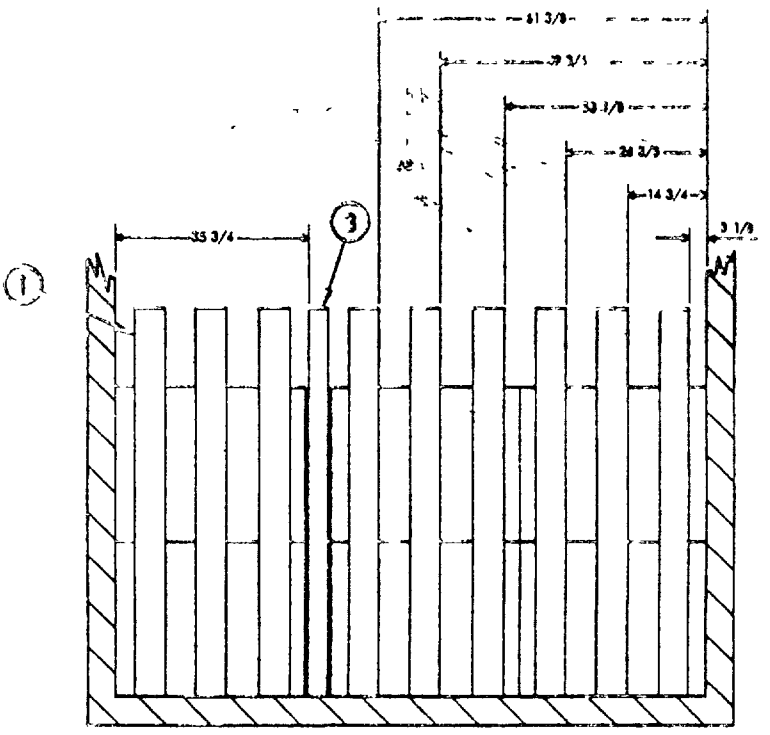
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



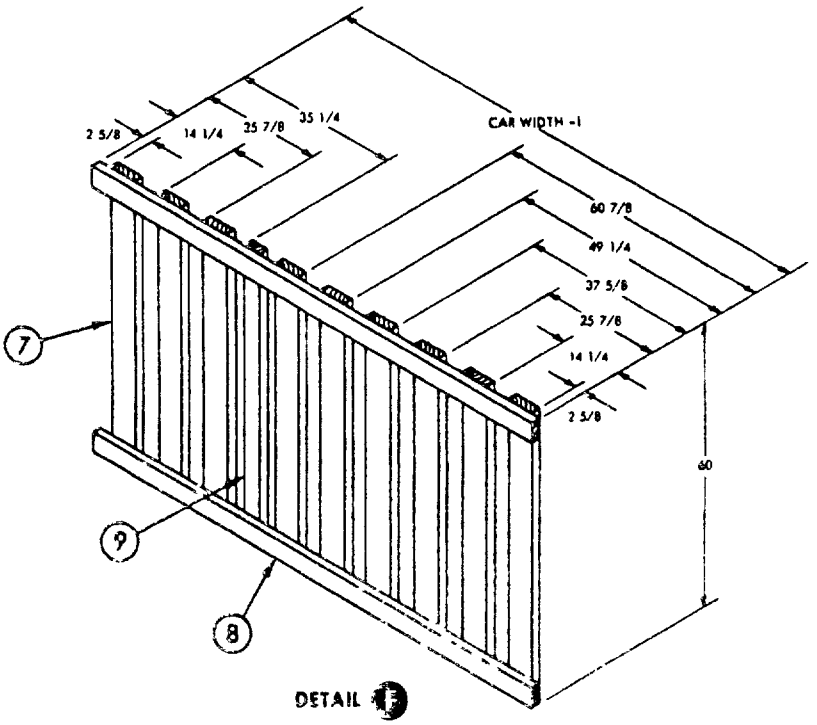
CARLOAD DATA

NUMBER OF UNIT LOADS	40
LOAD WEIGHT	123,760 LBS (EST.)
DUNNAGE WEIGHT	2,810 LBS
CARLOAD WEIGHT	126,570 LBS (EST.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



SECTION C-C
SHOWING LOCATION OF END WALL MEMBERS



DETAIL F
SEPARATOR GATE
3 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL

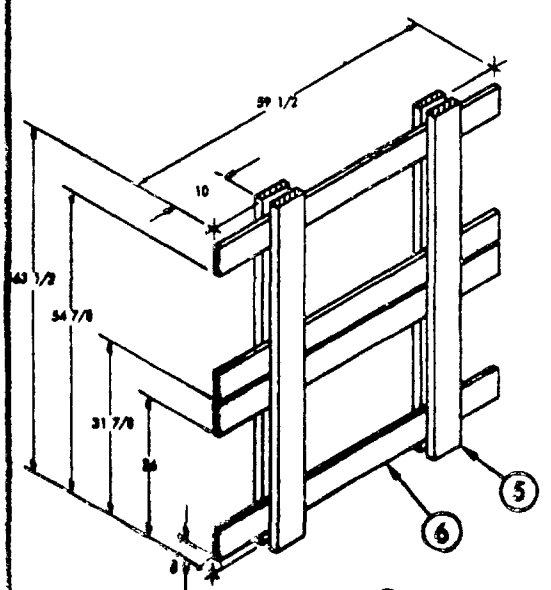
1 THE CARLOAD CONSISTS OF 48 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING

2 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100 SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED

3 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 17 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING

4 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 17 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE RANDOM LENGTH MATERIAL DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET MAY BE USED STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS DOORWAY PROTECTION IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER

5 WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT 2 IN USE ALTERNATE SWAY BRACE FRAMES SHOWN ON PAGE 38 IN PLACE OF DETAIL A



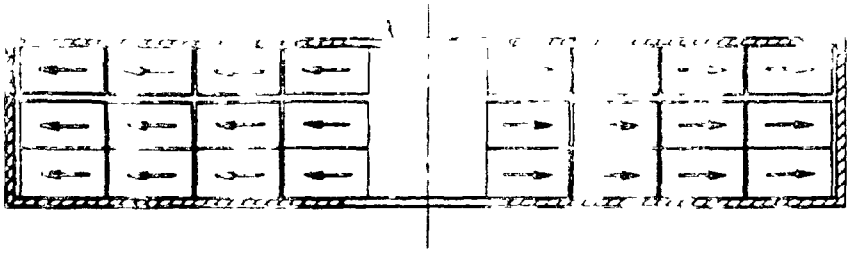
DETAIL A
SWAY BRACE FRAME
(3 REQUIRED)

* 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4 S

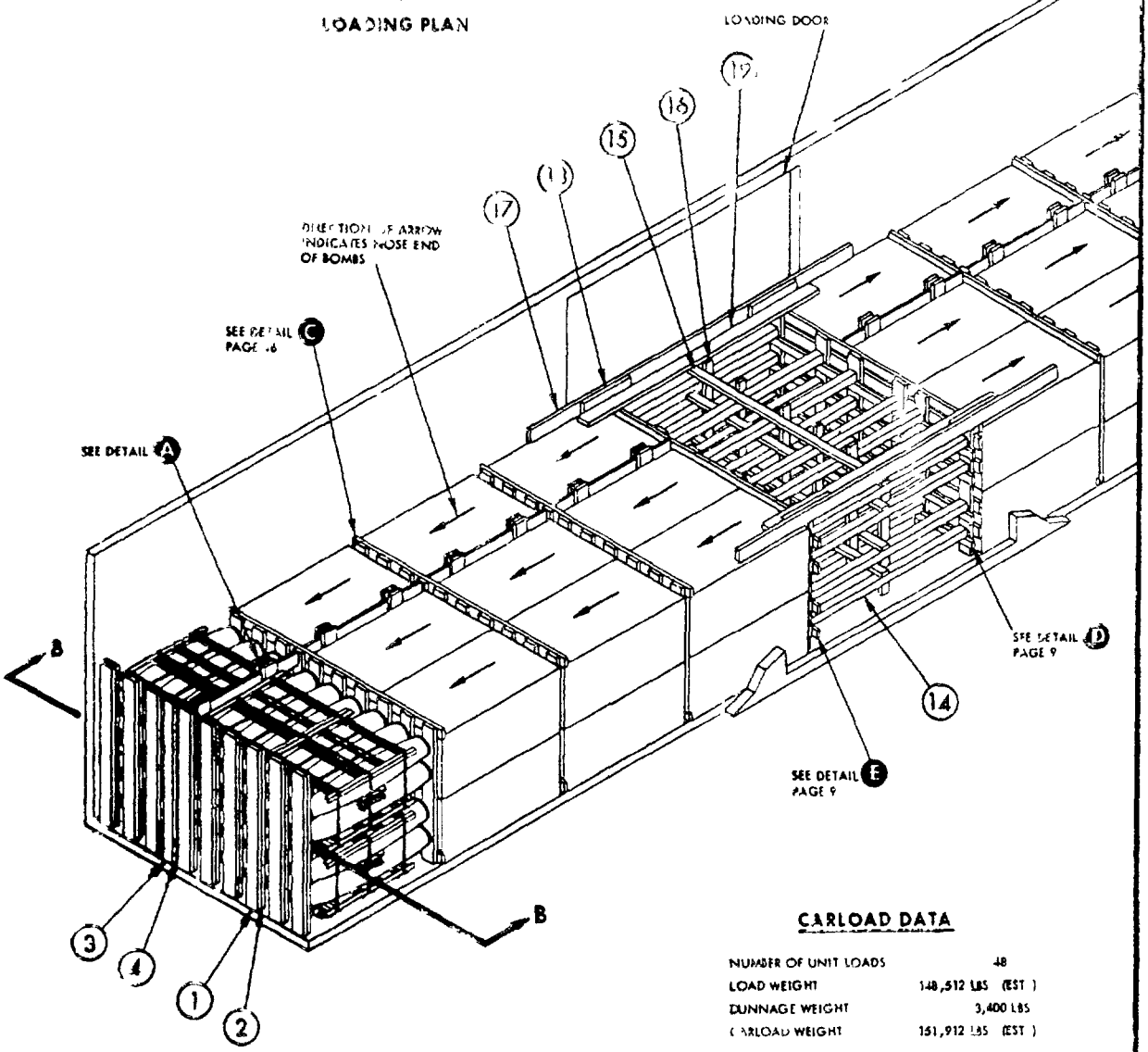
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE
19	CENTER GATE SWAY BRACE	2 x 6 x 10 FT	2	13	3 PER JOINT	16d
18	HOLD DOWN CLEAT	2 x 6 x 24	4	17	5	10d
17	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
16	VERTICAL TIE BAR	2 x 4 x 60	6	14	2 PER JOINT	16d
15	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH-1	4	14	2 PER JOINT	16d
14	STRUT	4 x 4 x WEDGE FIT *	24	11	2 PER JOINT	16d
13	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	11	3 PER JOINT	10d
12	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	1	3 PER JOINT	10d
11	CENTER GATE VERTICAL	2 x 6 x 60	12	SEE 10	-	-
10	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 60	6	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	12	7 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 60	54	SEE 8	-	-
6	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/4	32	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 63 1/2	32	6	3 PER JOINT	10d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



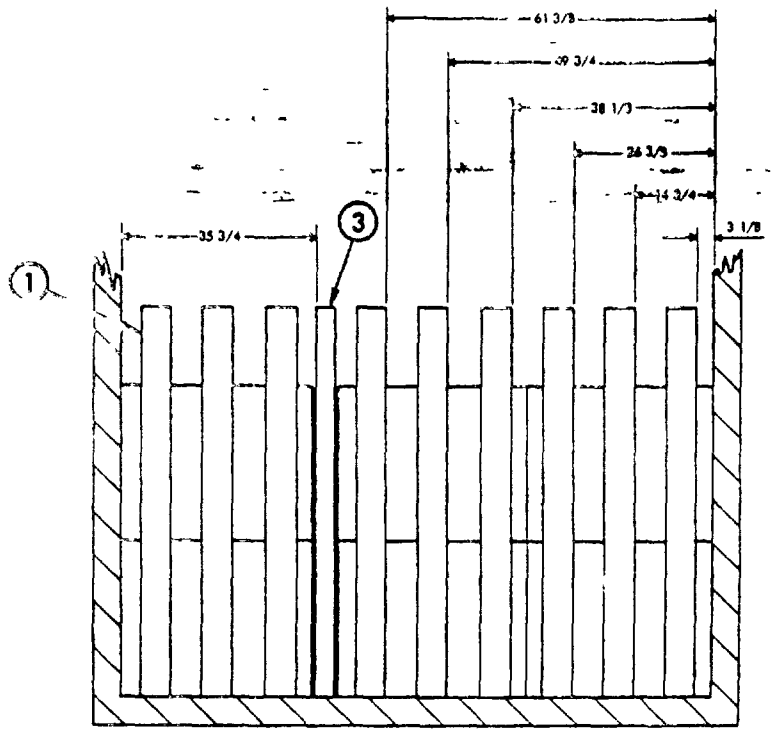
LOADING PLAN



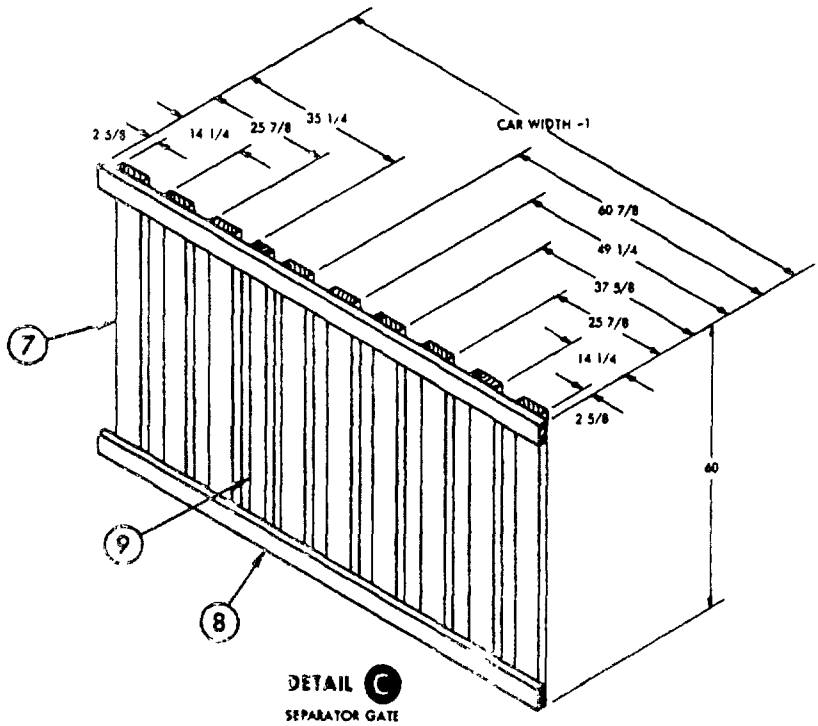
CARLOAD DATA

NUMBER OF UNIT LOADS	48
LOAD WEIGHT	148,512 LBS (EST.)
DUNNAGE WEIGHT	3,400 LBS
CARLOAD WEIGHT	151,912 LBS (EST.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



SECTION B-B
SHOWING LOCATION OF END WALL MEMBERS



DETAIL C
SEPARATOR GATE
6 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF EITHER 30 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING OR 27 UNIT LOADS COMPRISING THE FIRST LAYER ONLY OF THIS PROCEDURAL DRAWING
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT. WIDE PROVIDING THE LENGTH OF PIECE 29 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 29 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR WHETHER MAIN OR AUXILIARY EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK. EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND DOORWAY PROTECTION. PIECES 31 AND 32 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT. 2 IN. USE ALTERNATE SWAY BRACE SHOWN ON PAGE 38 IN PLACE OF DETAIL B AND C.

* 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's

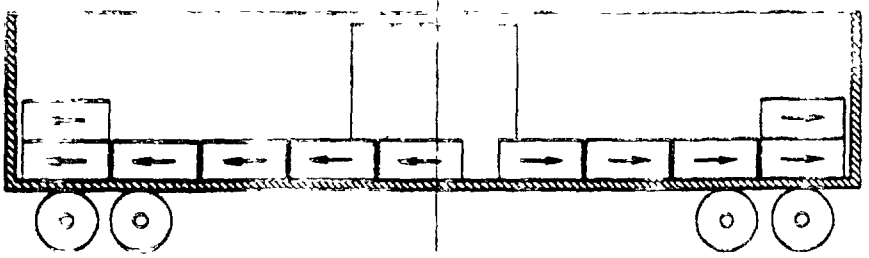
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
32	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	6	31	2 PER JOINT	10d
31	VERTICAL DOORWAY MEMBER	2 x 3 x 36	4	DOOR POST	2 PER FOOT	20d
30	HOLD DOWN CLEAT	2 x 6 x 60	2	29	2 PER FOOT	10d
29	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
28	STRUT	4 x 4 x WEDGE FIT *	12	16	2 PER JOINT	16d
27	HORIZONTAL BACK-UP CLEAT	2 x 6 x 30	8	24	14	16d
26	DIAGONAL BRACE	4 x 4 x 50	8	14,24	1 EACH END	60d
25	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	24	7	16d
24	HORIZONTAL WALL CLEAT	2 x 6 x 72	8	CAR WALL	16	10d
23	CENTER CLEAT	2 x 4 x 36	4	14	7	16d
22	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	4	10d
21	VERTICAL BACK-UP CLEAT	2 x 6 x 36	4	CAR WALL	8	10d
20	INTERMEDIATE WALL CLEAT	2 x 4 x 7 3/4	4	CAR WALL	2	10d
19	SWAY BRACE STOP	2 x 6 x 22	2	16	3 PER JOINT	10d
18	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	16	5 PER JOINT	10d
17	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	4	16	3 PER JOINT	10d

LIST OF MATERIALS AND NAILING DATA

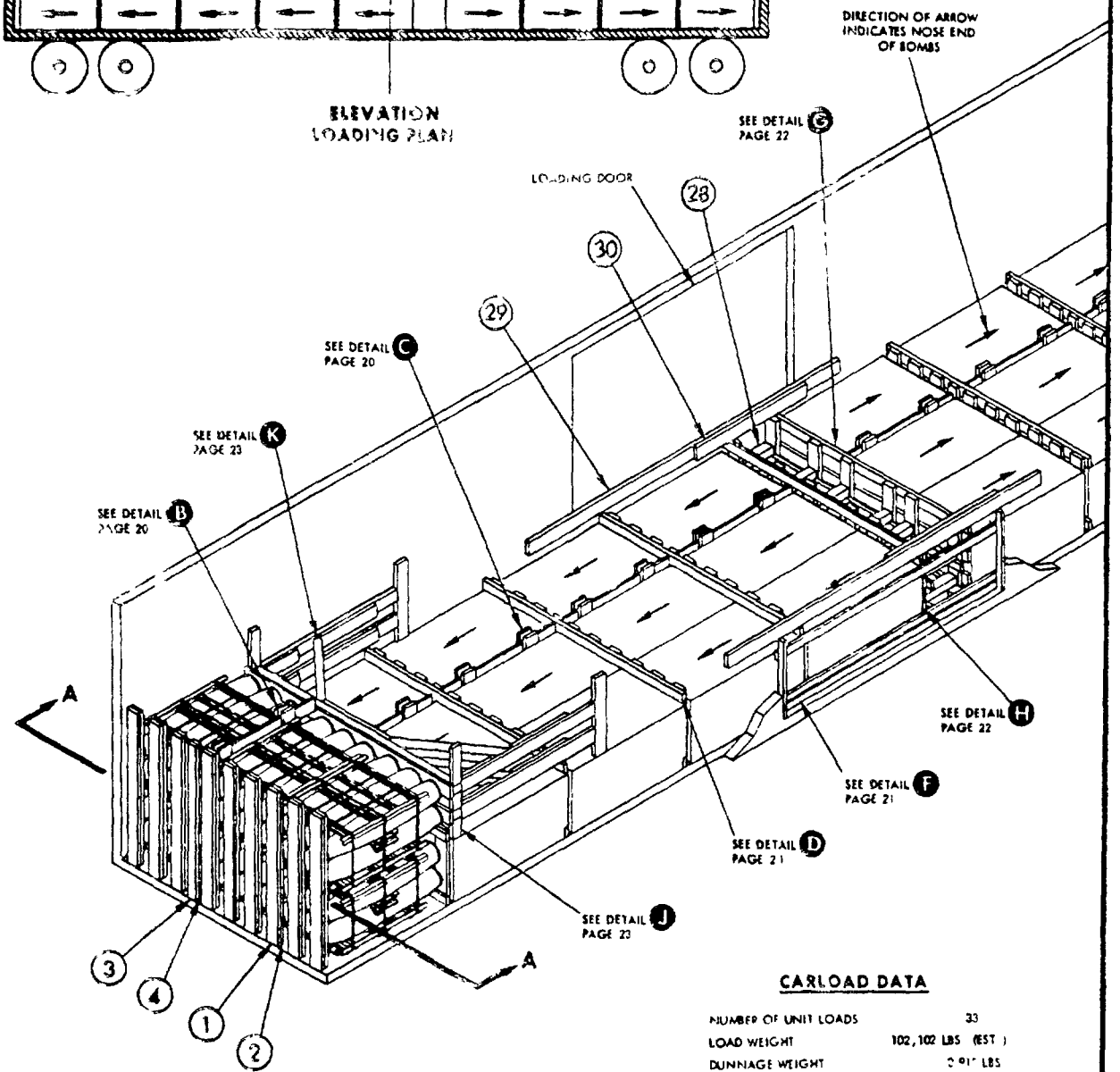
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
16	CENTER GATE VERTICAL	2 x 6 x 36	12	SEE 15	-	-
15	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	4	16	3 PER JOINT	10d
14	CROSS BRACE	4 x 4 x CAR WIDTH	4	SEE 13	-	-
13	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH	4	14	2 PER FOOT	16d
12	LOWER WALL CLEAT	2 x 4 x 9	4	CAR WALL	3	10d
11	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/2	14	SEE 10	-	-
10	SWAY BRACE VERTICAL	2 x 6 x 34 1/2	28	11	3 PER JOINT	10d
9	SWAY BRACE STOP	2 x 4 x 34	7	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	14	7, 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 34	63	SEE 8	-	-
6	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/2	8	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 63 1/2	8	6	3 PER JOINT	10d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



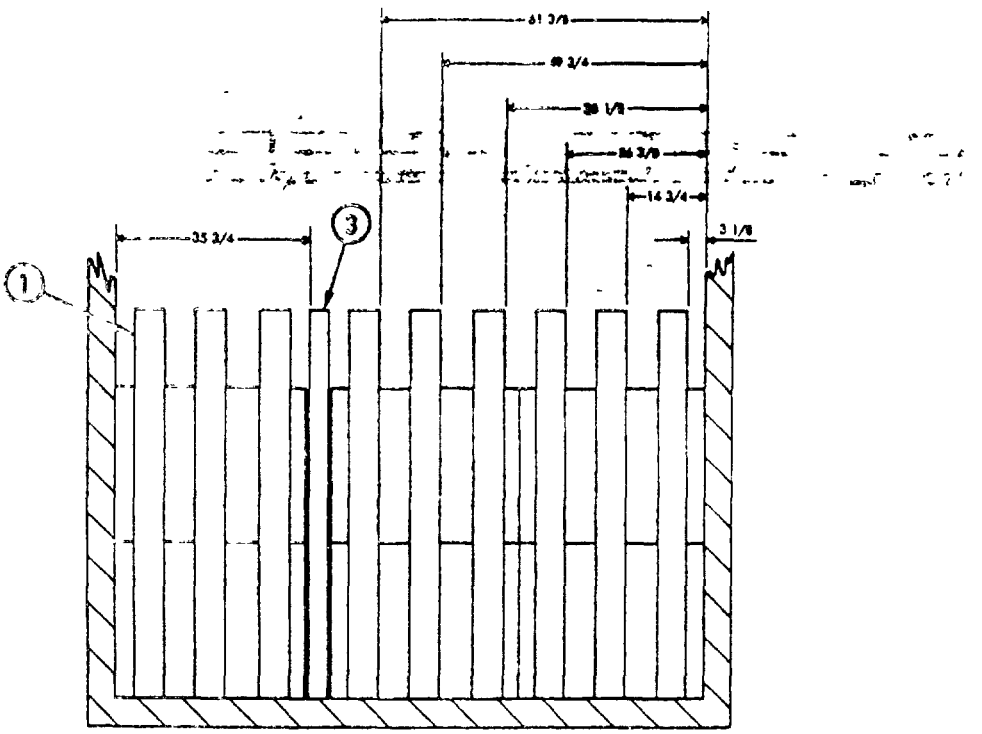
ELEVATION
LOADING PLAN



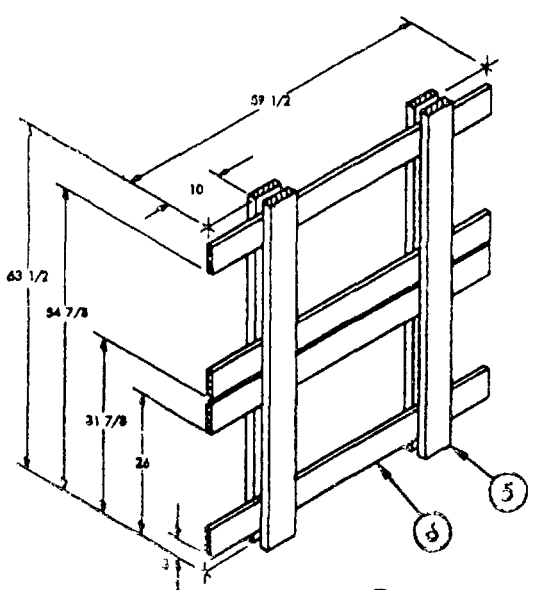
CARLOAD DATA

NUMBER OF UNIT LOADS	33
LOAD WEIGHT	102,102 LBS (EST.)
DUNNAGE WEIGHT	2,918 LBS
CARLOAD WEIGHT	105,020 LBS (EST.)

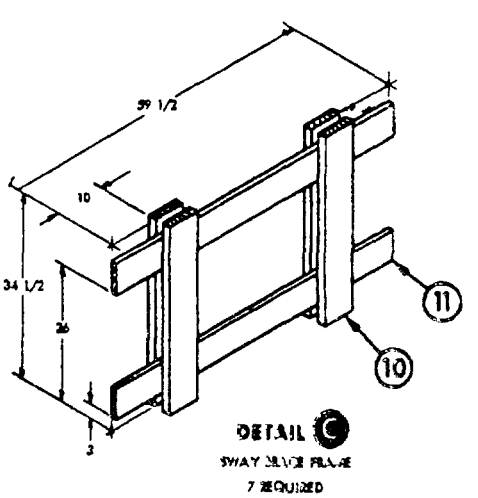
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



SECTION A-A
SHOWING LOCATION OF END WALL MEMBERS

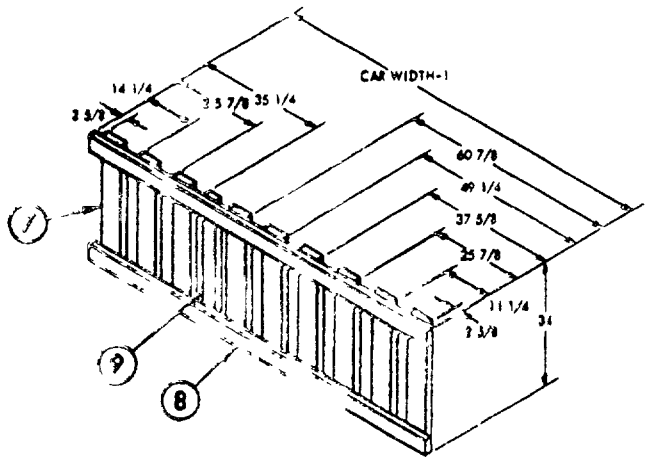


DETAIL B
SWAY BRACE FRAME
2 REQUIRED

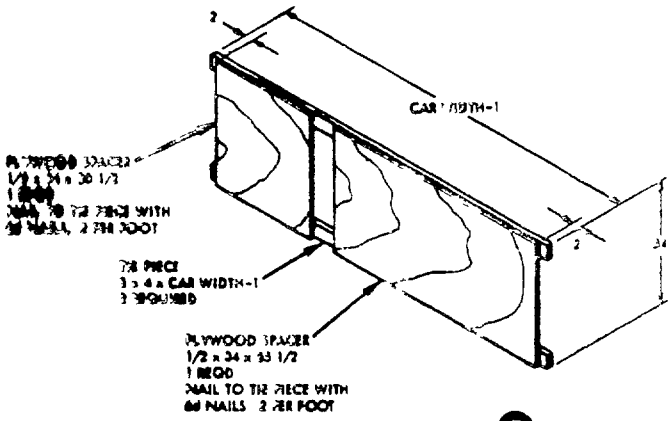


DETAIL C
SWAY BRACE FRAME
7 REQUIRED

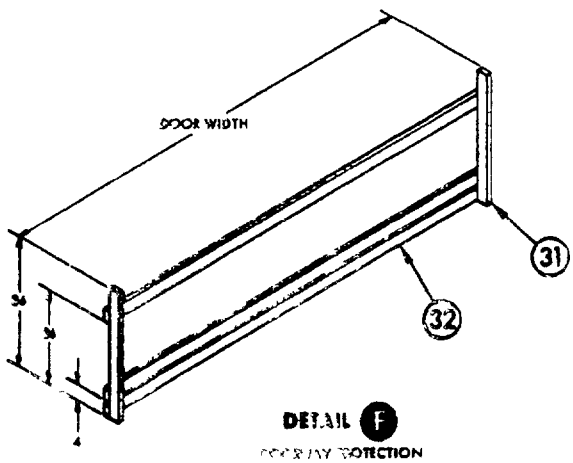
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



DETAIL D
SEPARATOR GATE
7 REQUIRED



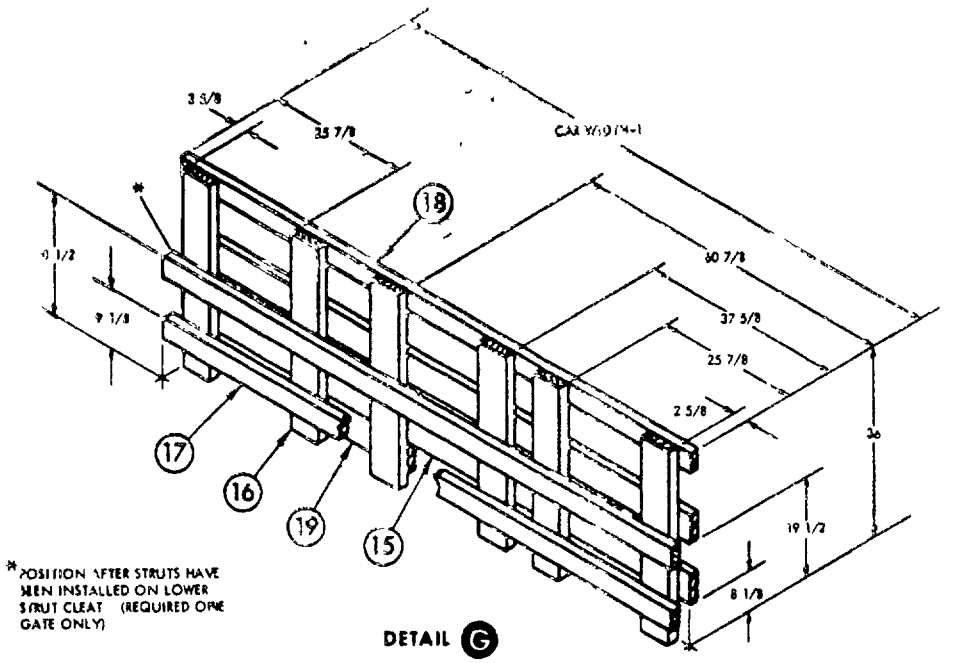
DETAIL E
ALTERNATE SEPARATOR GATE
7 REQUIRED



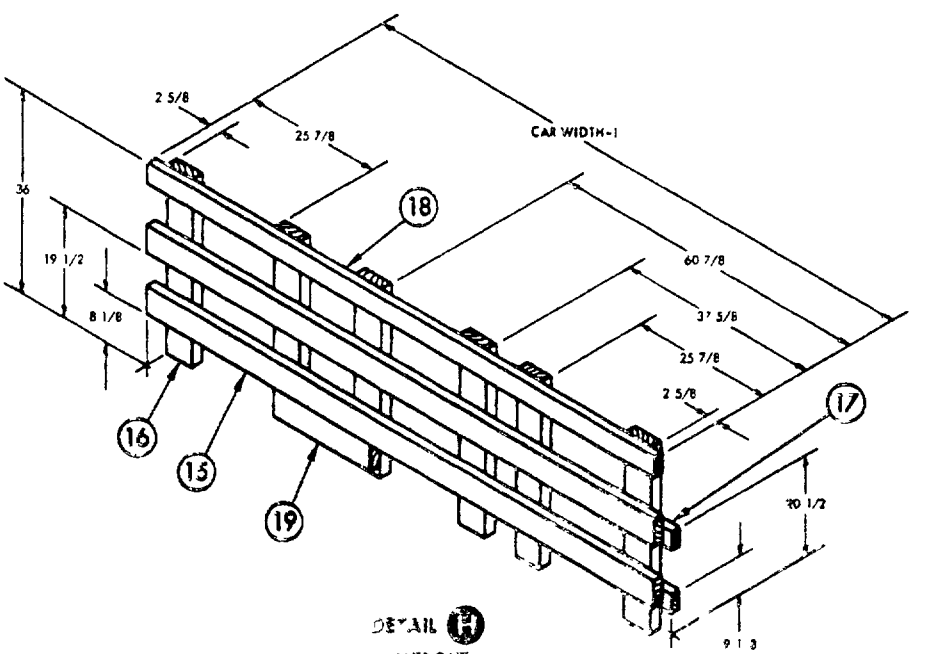
WHEN BOXCAR HAS STEEL DOOR POSTS
DOORWAY PROTECTION AS PER MIL-STD-1125
(NAVY) PAGE 36, FIG 21 IS REQUIRED

DETAIL F
DOORWAY PROTECTION
2 REQD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

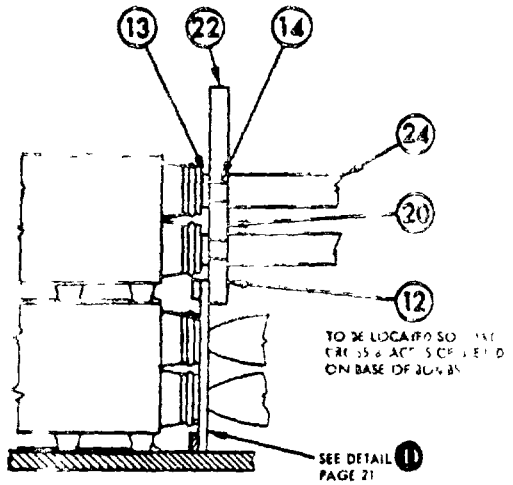


DETAIL G
CENTER GATE
1 REQUIRED

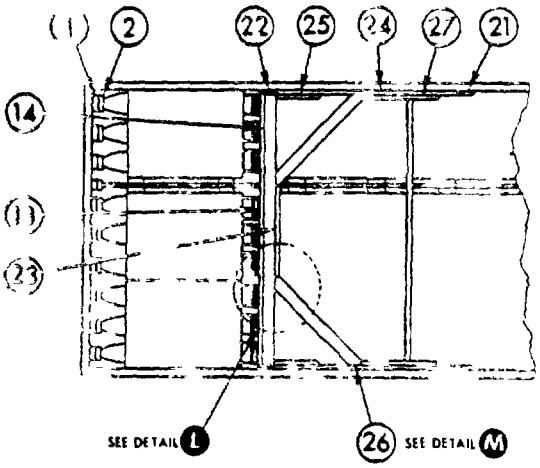


DETAIL H
CENTER GATE
1 REQUIRED

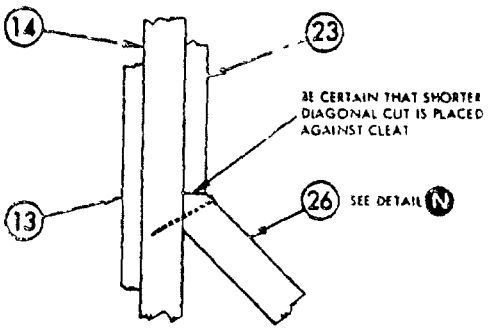
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



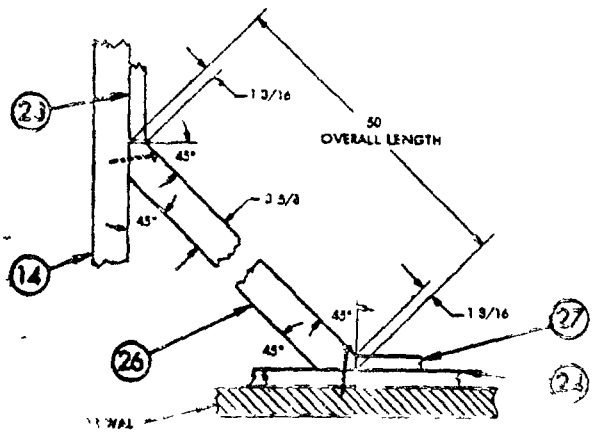
DETAIL J



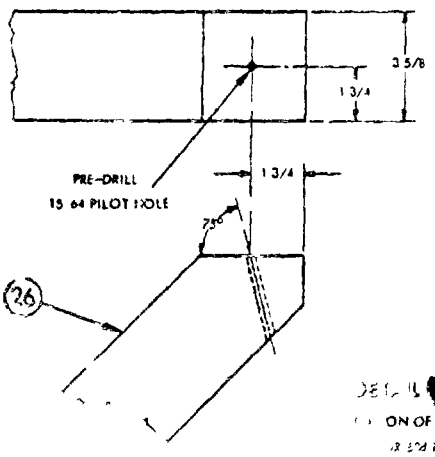
DETAIL K
PARTIAL LAYER BRACING



DETAIL L
POSITIONING OF DIAGONAL BRACE



DETAIL M
POSITIONING OF
DIAGONAL BRACE

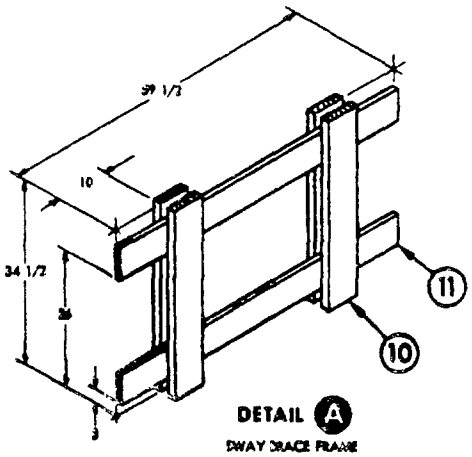


DETAIL N
ON OF PILOT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF EITHER 30 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, OR 39 UNIT LOADS IN ACCORDANCE WITH THE ALTERNATE PLAN AND DETAILS DEPICTED ON PAGES 28 & 29 OF THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 21 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 21 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND DOORWAY PROTECTION, PIECES 23 AND 24, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT 2 IN. USE ALTERNATE SWAY BRACE FRAMES SHOWN ON PAGE 38 IN PLACE OF DETAIL A AND C.



DETAIL A

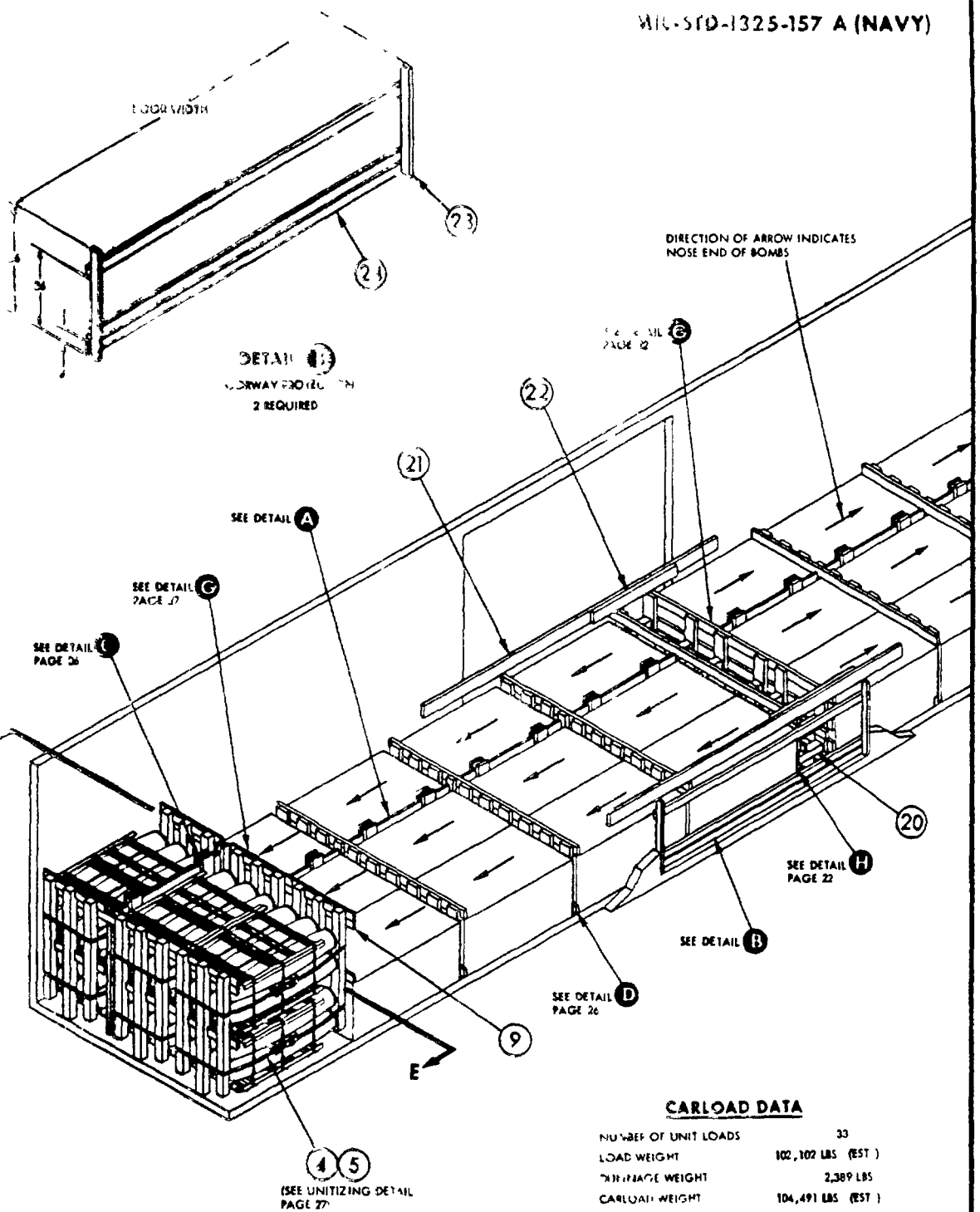
SWAY BRACE FRAME

7 REQUIRED

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NO PER NAILS	SIZE
25	SPACER	2 x 4 x 8 1/4	48	3	3	10d
24	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	6	23	2 PER JOINT	10
23	VERTICAL DOORWAY MEMBER	2 x 3 x 36	4	DOOR POST	2 PER FOOT	20d
22	HOLD DOWN CLEAT	2 x 6 x 48	2	21	2 PER FOOT	10d
21	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
20	STRUT	4 x 4 x WEDGE FIT	12	16	2 PER JOINT	16d
19	SWAY BRACE STOP	2 x 6 x 22	2	16	3 PER JOINT	10d
18	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	16	3 PER JOINT	10d
17	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	4	16	3 PER JOINT	10d
16	CENTER GATE VERTICAL	2 x 6 x 36	12	SEE 15	-	-
15	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	4	16	3 PER JOINT	10d
14	SWAY BRACE STOP	2 x 4 x 34	5	SEE 13	-	-
13	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	10	12 14	3 PER JOINT	10d
12	SEPARATOR GATE VERTICAL	2 x 6 x 34	45	SEE 13	-	-
11	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/2	14	SEE 10	-	-
10	SWAY BRACE VERTICAL	2 x 6 x 34 1/2	28	11	3 PER JOINT	10d
9	TIE PIECE	2 x 4 x CAR WIDTH-1	4	1, 2	1 PER JOINT	16d
8	SWAY BRACE HORIZONTAL	1 x 6 x 63	8	SEE 6	-	-
7	SWAY BRACE VERTICAL	1 x 6 x 43 1/2	8	8	3 PER JOINT	10d
6	SWAY BRACE VERTICAL	2 x 4 x 63 1/2	8	8	3 PER JOINT	10d
5	SEAL	1 1 4	24	-	-	-
4	STRAP	1 1/4 x .035 19 FT.	1	-	-	-
3	TIE PIECE	2 x 4 x 35	24	1, 2	2 PER JOINT	16d
2	UPRIGHT	4 x 4 x 59	24	SEE 3	-	-
1	UPRIGHT	4 x 4 x 64	12	SEE 3	-	-

LIST OF MATERIALS AND NAILING DATA

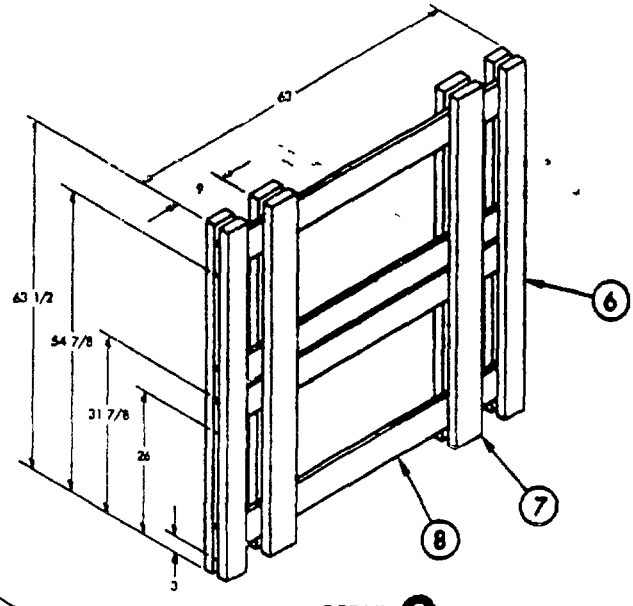
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



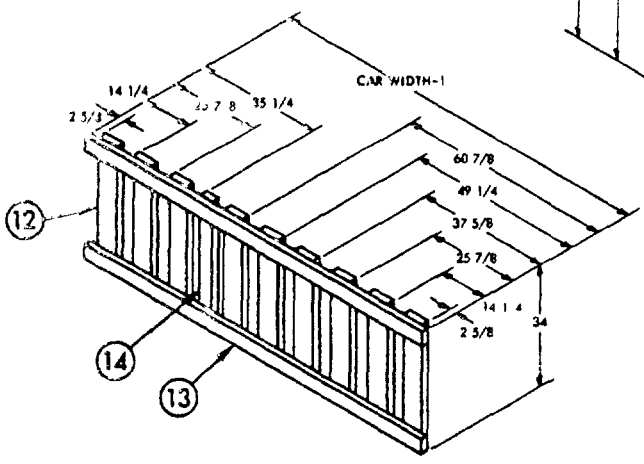
CARLOAD DATA

NUMBER OF UNIT LOADS	33
LOAD WEIGHT	102,102 LBS (EST.)
DUNNAGE WEIGHT	2,389 LBS
CARLOAD WEIGHT	104,491 LBS (EST.)

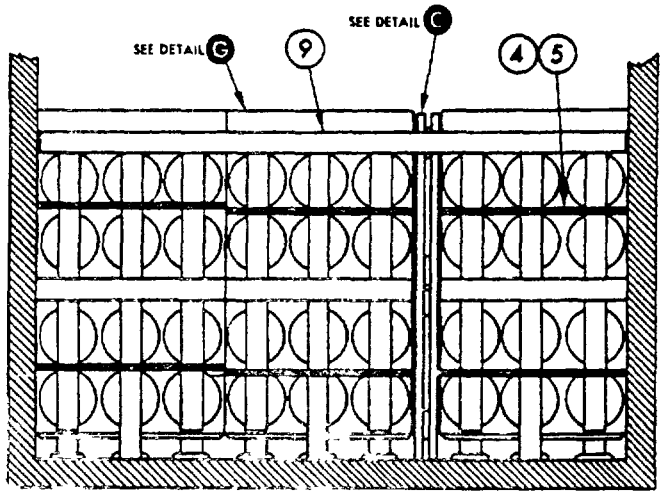
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



DETAIL C
SWAY BRACE
2 REQUIRED

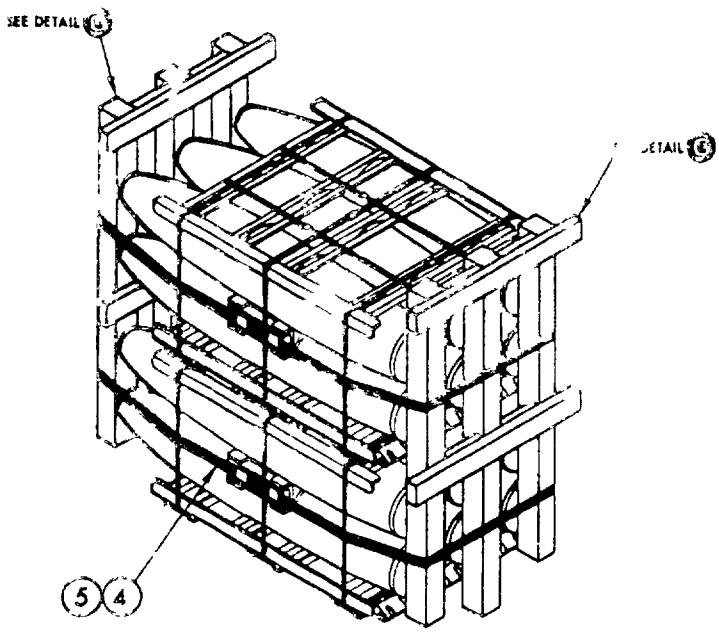


DETAIL D
SEPARATOR GATE
5 REQUIRED
(SEE ALTERNATE SEPARATOR GATE DETAIL E PAGE 21)



SECTION E-E

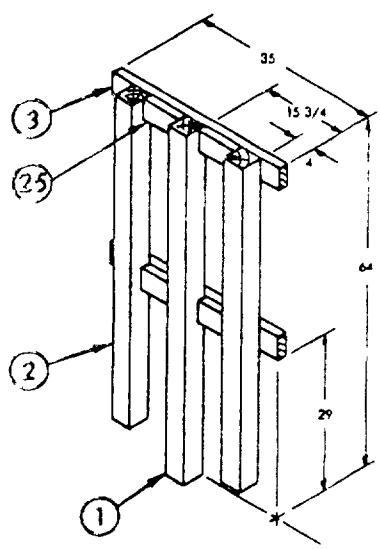
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



DETAIL F

UNITIZING DETAIL

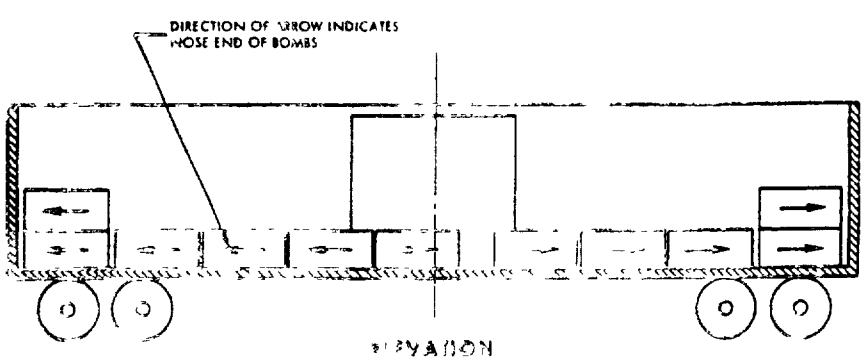
TE ..ION AND SEAL WITH TWO DOUBLE CRIMPED SEALS PER STRAP



DETAIL G

PARTIAL LAYER BLOCKING

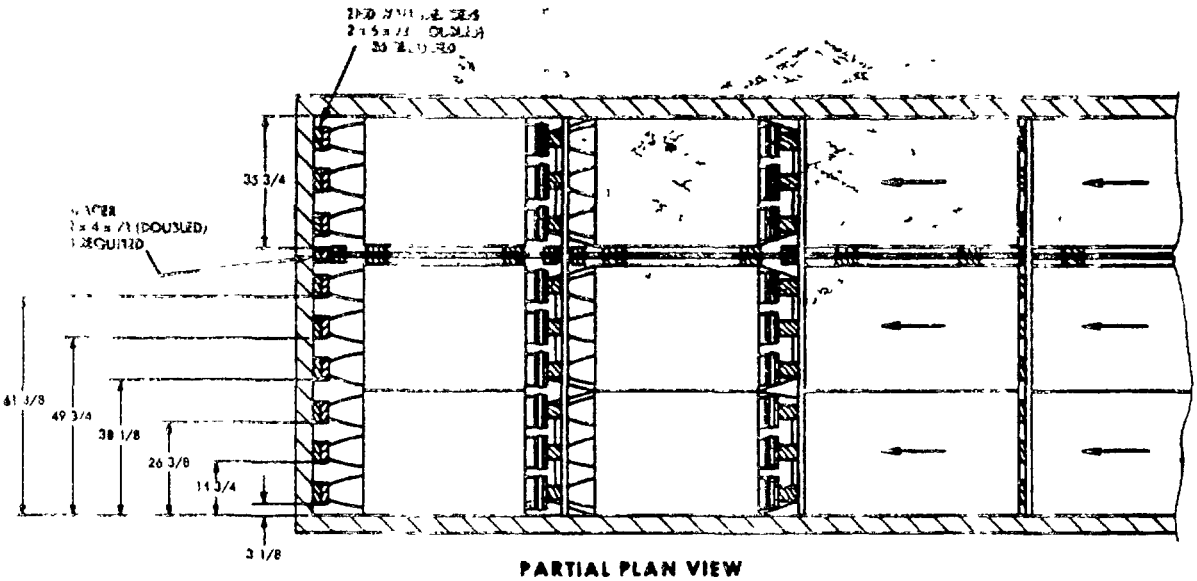
12 REQUIRED



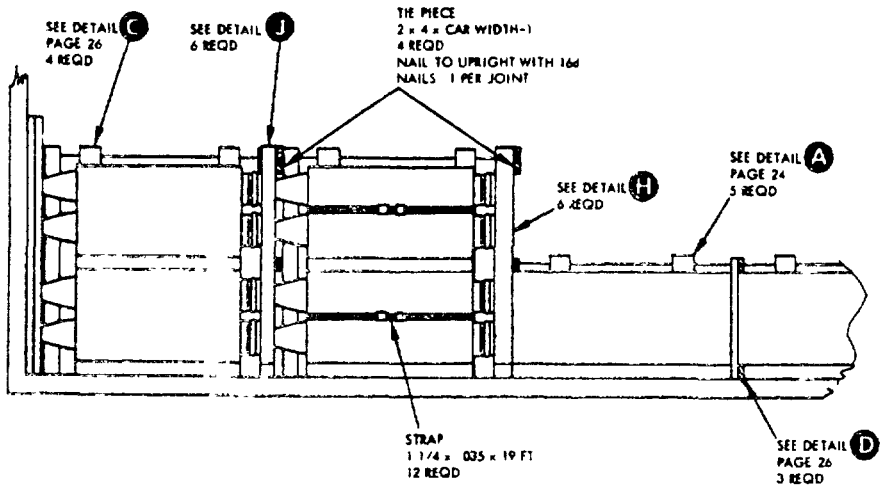
ELEVATION

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

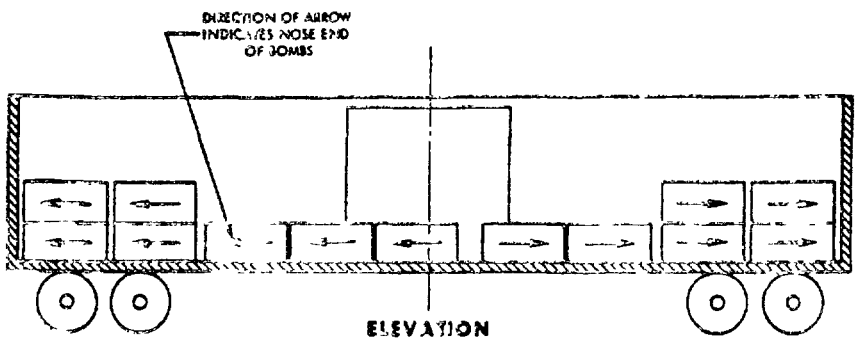
ALTERNATE PLAN FOR 39 UNIT LOADS



PARTIAL PLAN VIEW

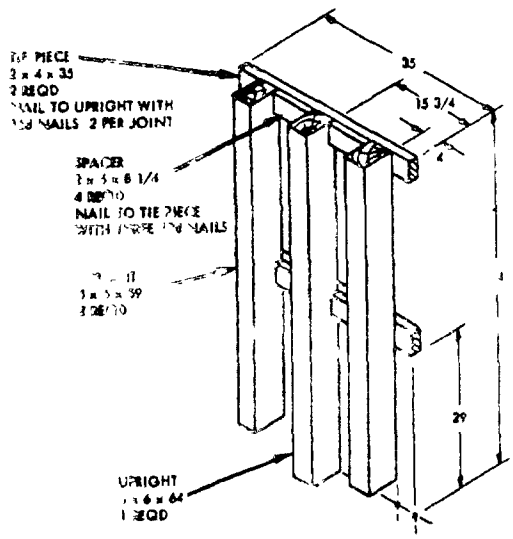


PARTIAL ELEVATION

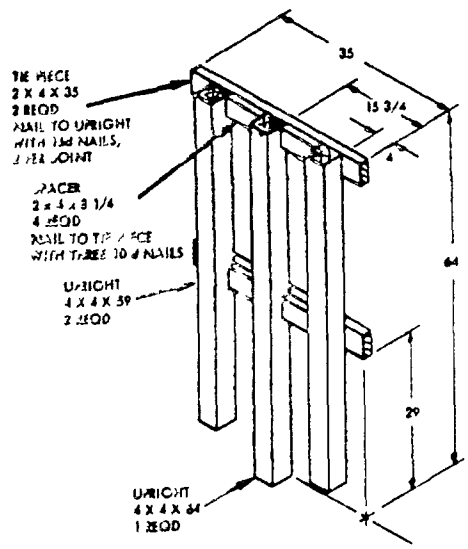


ELEVATION
LOADING PLAN

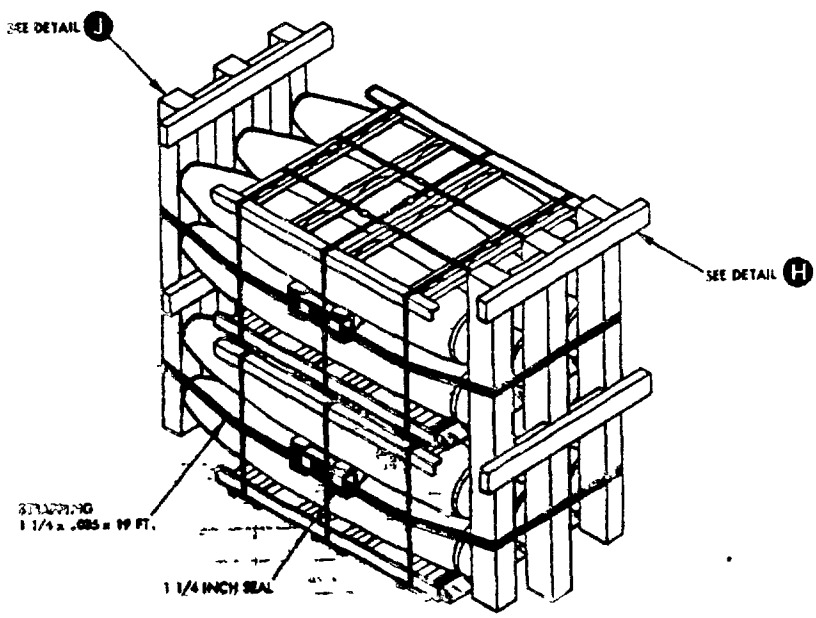
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



DETAIL H
PARTIAL LAYER BLOCKING
6 REQUIRED



DETAIL J
PARTIAL LAYER BLOCKING
6 REQUIRED



DETAIL K
UNITIZING SEAL
RAISED SEAL WITH TWO DOUBLE
CRIMPED SEALS 3 STRAP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

50 FT 6 IN. BOXCAR, COMMERCIAL (USING LOAD DIVIDERS)

1 THESE PROCEDURES DEPICT THE METHOD OF LOADING THE FLEET ISSUE UNIT LOAD OF MK 82 BOMBS (THERMALLY PROTECTED) IN CUSHIONED BOXCARS WHICH ARE EQUIPPED WITH LOAD DIVIDERS AND WITH OR WITHOUT ADJUSTABLE SIDE FILLERS ONLY CUSHIONED CARS THAT HAVE SLIDING CENTER SILL TYPE CUSHIONING DEVICES OR END-OF-CAR TYPE DEVICES WHICH HAVE AT LEAST 15 INCHES OF TRAVEL ARE ACCEPTABLE (ONLY THOSE CUSHIONED BOXCARS EQUIPPED WITH LOAD DIVIDERS MANUFACTURED BY EVANS, EQUIPCO AND PRECO HAVE BEEN TESTED AND APPROVED BY THE AAR, BUX) THE LOAD DIVIDERS WILL REPLACE THE CENTER GATE ASSEMBLY A STRUT ASSEMBLY, DETAIL G, WILL BE REQUIRED BETWEEN THE LOAD DIVIDERS AS SHOWN IN THE LOAD VIEWS, IF EITHER LOAD DIVIDER IS REQUIRED TO RETAIN A LADING WEIGHT OF 50,000 POUNDS OR MORE

2 BOXCARS EQUIPPED WITH ADJUSTABLE SIDE FILLERS THAT HAVE 3/8 INCH OR THICKER PANELS MAY BE USED HOWEVER, THESE SIDE FILLERS MUST NOT BE USED FOR LATERAL BLOCKING THEY MUST BE RETRACTED AND LOCKED AGAINST THE CAR SIDE WALL, AND A FILL PIECE INSTALLED IN THE VOID, IF ONE EXISTS, BETWEEN THE CAR SIDE WALL AND THE SIDE FILLER PANEL (SEE DETAIL H, PAGE 34)

3 PRIOR TO LOADING A CAR EQUIPPED WITH LOAD DIVIDERS, A VERY CAREFUL INSPECTION MUST BE MADE TO ENSURE THAT THE CAR AND THE CAR EQUIPMENT IS IN GOOD CONDITION THE CONDITION OF THE LOAD DIVIDERS SHOULD BE CHECKED THOROUGHLY BREAKS IN WELDING BENT OR OTHERWISE DAMAGED LOCKING PINS, AND BENT OR DEFORMED BULKHEADS ARE REASONS FOR REJECTING CARS AFTER THE LOAD DIVIDERS ARE POSITIONED AGAINST THE LADING AND THE LOCKING PINS ARE ENGAGED IN THE HOLES OF THE RAILS, THE LOCKING PINS MUST BE INSPECTED TO ENSURE THAT THE PINS ARE FULLY ENGAGED IN THE LOCKING HOLES IF THE LOWER PINS ARE NOT FULLY SEATED IN THE HOLES, THE LINKAGE MECHANISM SHOULD BE ADJUSTED AS REQUIRED SO THAT THE PINS WILL BE FULLY SEATED INTO THE LOCKING HOLES OF THE LOWER RAILS IF PRESENT, DEBRIS MUST BE REMOVED FROM BENEATH THE LOCKING HOLES SELECTED FOR SECURING A DIVIDER

4 THE CARLOAD CONSISTS OF 46 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING THERE ARE 23 UNIT LOADS AND ONE DUMMY UNIT LOAD IN EACH END OF THE CAR THE DUMMY UNIT LOAD IS REQUIRED TO FILL OUT THE VOID SPACE AND MUST BE CONSTRUCTED AS SHOWN ON PAGE 36 (IF LOAD LIMIT OF CAR PERMITS, REPLACE DUMMY LOADS WITH BOMB LOADS)

5 WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS ENTIRE STACKS SHALL BE ELIMINATED WHERE POSSIBLE, OTHERWISE ADDITIONAL DUMMY UNIT LOADS SHALL BE USED TO FILL OUT THE VOIDS IN A STACK

6 THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING DOORWAY PROTECTION IS INSTALLED WHEN REQUIRED (SEE DETAIL B, PAGE 11)

7 THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR WHETHER MAIN OR AUXILIARY EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4 INCH STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS DOORWAY PROTECTION IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER

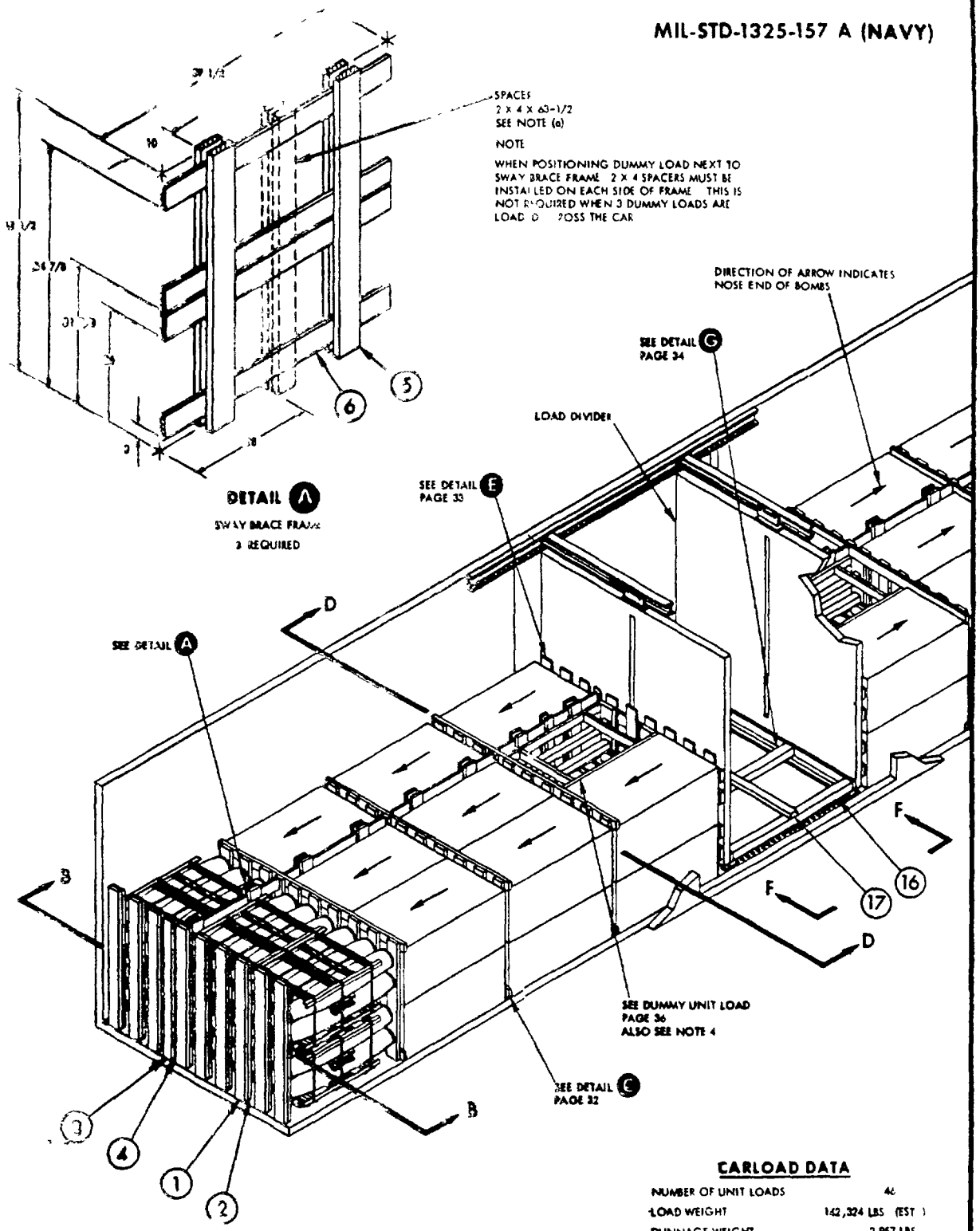
8 WHEN LOADING BOXCARS WITH AN INSIDE WIDTH GREATER THAN 9 FT 2 IN USE ALTERNATE SWAY BRACE FRAMES SHOWN ON PAGE 38 IN PLACE OF DETAIL A

*"W" = DISTANCE BETWEEN LOCKING PINS AT EACH SIDE OF THE LOAD DIVIDER

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
17	TIE BAR	2 x 4 x CAR WIDTH-1	1	16	3 PER JOINT	10d
16	STRUT	4 x 4 CUT TO FIT	4	15	2 EACH END	16d
15	BUFFER PIECE	2 x 4 x CAR WIDTH-1	2	SEE 14	-	-
14	STRUT LEDGER	2 x 4 x CAR WIDTH-1	2	15	1 PER FOOT	10d
13	HOLD DOWN	1 x 6 x "W"*	2	14	1 PER FOOT	6d
12	SWAY BRACE STOP	1 x 4 x 61 7/8	2	11	3 PER JOINT	6d
11	SEPARATOR GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	SEE 10	-	-
10	SEPARATOR GATE VERTICAL	1 x 6 x 61 7/8	18	11	3 PER JOINT	6d
9	SWAY BRACE STOP	2 x 4 x 60	6	SEE 8	-	-
8	SEPARATOR GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	12	7, 9	3 PER JOINT	10d
7	SEPARATOR GATE VERTICAL	2 x 6 x 60	54	SEE 8	-	-
6	SWAY BRACE HORIZONTAL	1 x 6 x 59 1/2	32	SEE 5	-	-
5	SWAY BRACE VERTICAL	2 x 6 x 63 1/2	32	6	3 PER JOINT	10d
4	END WALL MEMBER	2 x 4 x 72	2	3	1 PER FOOT	10d
3	END WALL MEMBER	2 x 4 x 72	2	CAR WALL	1 PER FOOT	10d
2	END WALL MEMBER	2 x 6 x 72	18	1	1 PER FOOT	10d
1	END WALL MEMBER	2 x 6 x 72	18	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS AND NAILING DATA

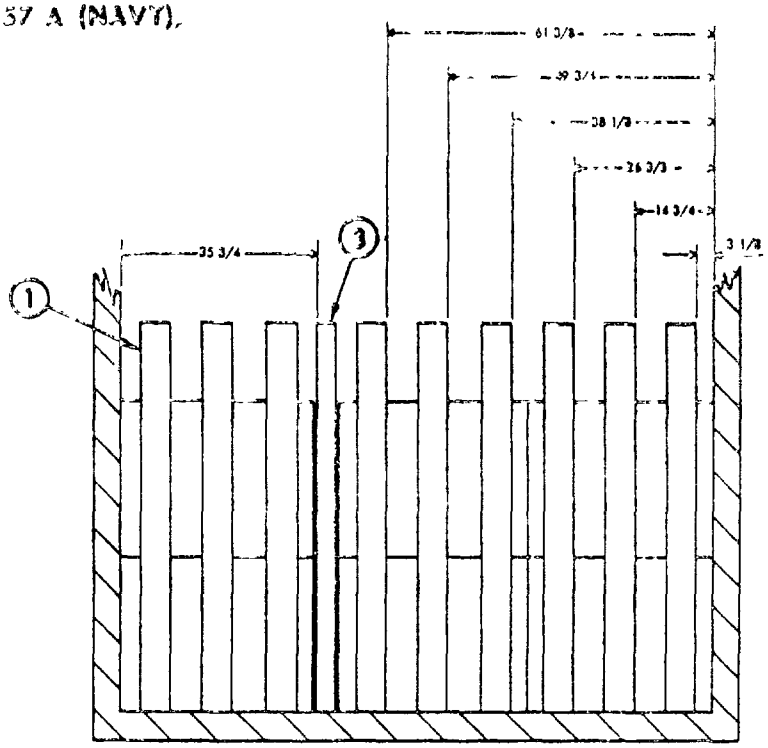
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



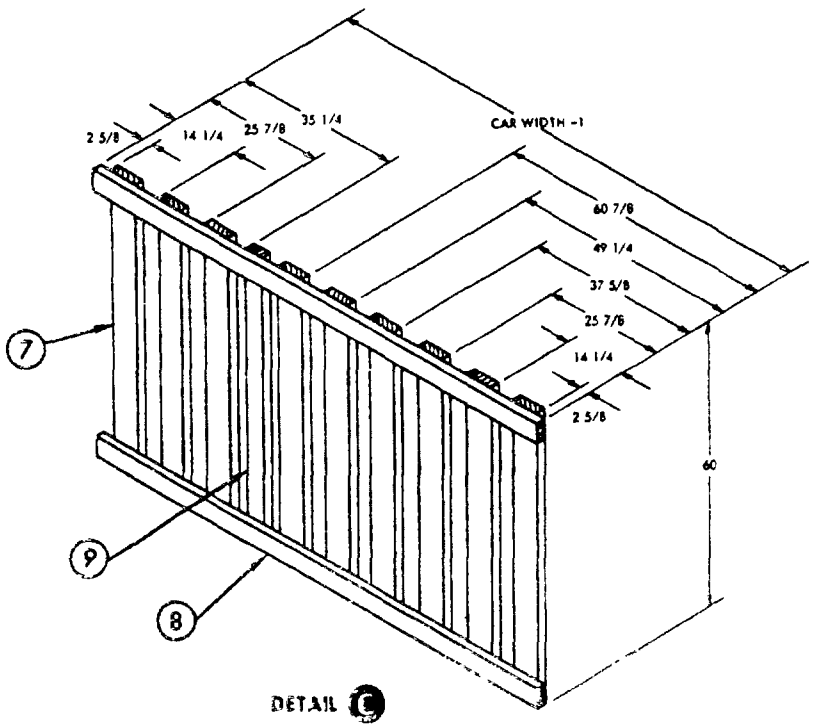
CARLOAD DATA

NUMBER OF UNIT LOADS	46
LOAD WEIGHT	142,324 LBS (EST.)
DUNNAGE WEIGHT	2 957 LBS
CARLOAD WEIGHT	145,281 LBS (EST.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

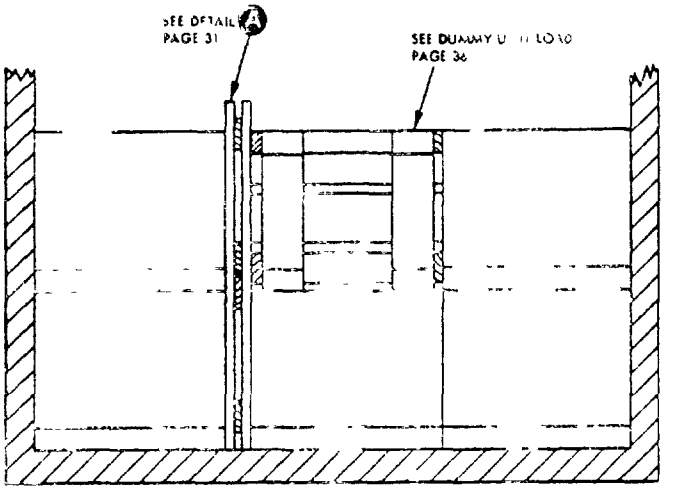


SECTION B-B
SHOWING LOCATION OF END WALL MEMBERS

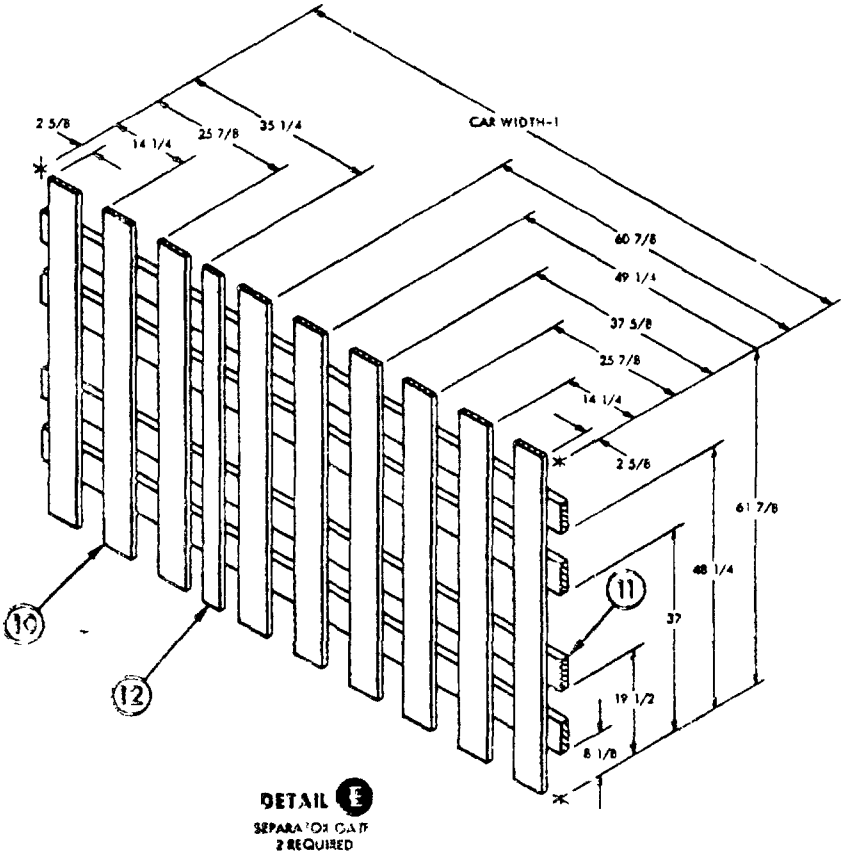


DETAIL C
SEPARATOR GATE
6 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

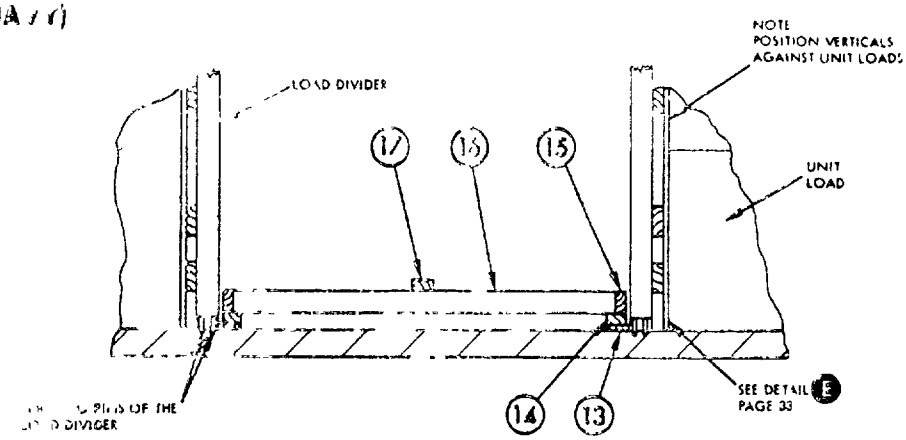


SECTION D-D

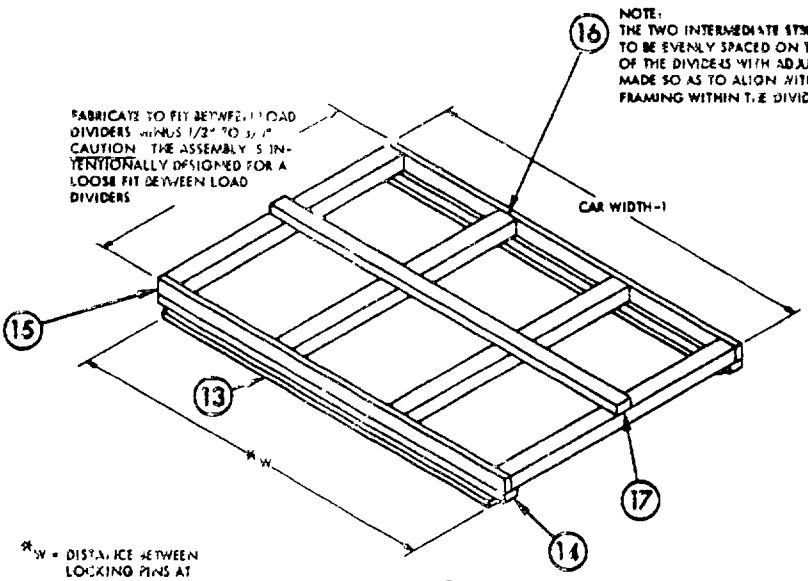


DETAIL E
SEPARATOR GATE
2 REQUIRED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

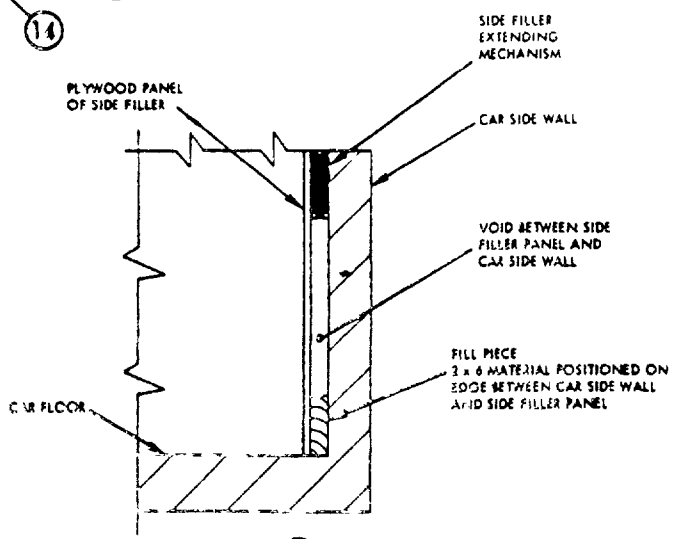


VIEW F-F
INSTALLATION OF STRUT ASSEMBLY



*W = DISTANCE BETWEEN LOCKING PINS AT EACH SIDE OF THE LOAD DIVIDER

DETAIL G
STRUT ASSEMBLY
1 REQUIRED



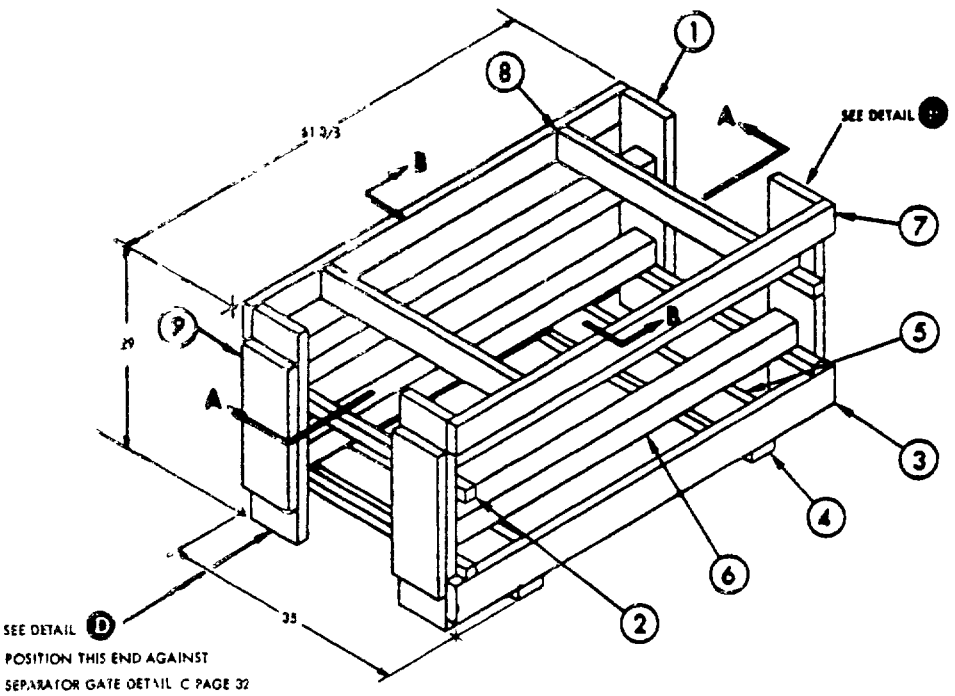
DETAIL H
SHOWING ARRANGEMENT FOR
ADJUSTABLE SIDE FILLERS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

DUMMY UNIT LOAD



NOTES

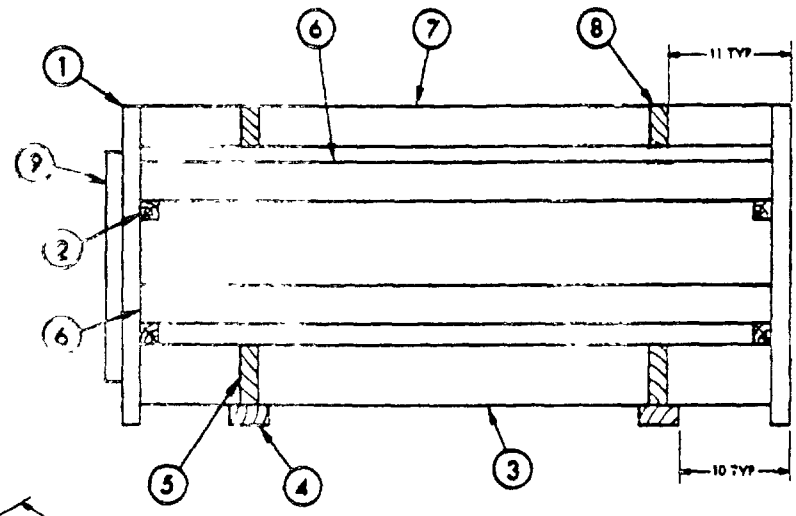
(1) NAIL SUPPORT (PIECE 4) TO LONGITUDINAL (PIECE 3) WITH 16d NAILS 2 PER JOINT AND TO CROSS BRACE (PIECE 5) WITH 5-16d NAILS

(2) TOENAIL STRUT TO VERTICAL WITH 16d NAILS 3 EACH END (TOP AND BOTH SIDES), ALSO END NAIL THROUGH VERTICAL WITH 16d NAILS 2 EACH END

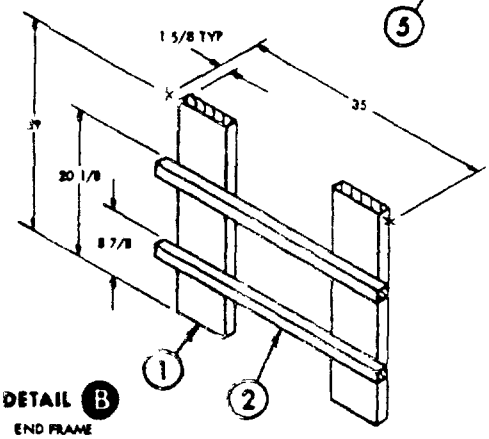
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
9	VERTICAL CLEAT	2 x 8 x 21	2	1	4	10d
8	CROSS BRACE	2 x 4 x CUT TO FIT	2	SEE 7	-	-
7	LONGITUDINAL	2 x 4 x 59 7/8	2	1 8	2 PER JOINT	16d
6	STRUT	4 x 4 x CUT TO FIT	4	1	SEE NOTE (2)	
5	CROSS BRACE	2 x 6 x CUT TO FIT	2	SEE 3	-	-
4	SUPPORT	2 x 4 x 35	2	3 5	SEE NOTE (1)	
3	LONGITUDINAL	2 x 6 x 59 7 8	2	1 5	3 PER JOINT	16d
2	STRUT CLEAT	2 x 2 x 35	4	1	3 PER JOINT	16d CLNHD
1	VERTICAL	2 x 8 x 29	4	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

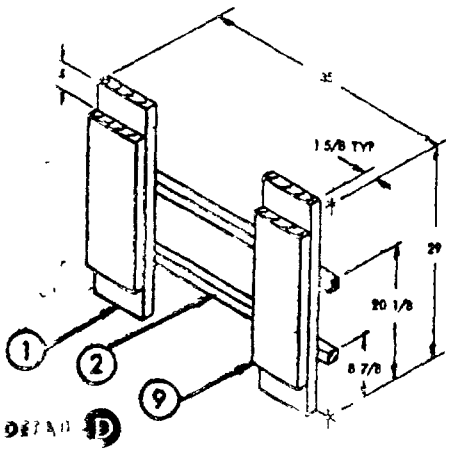
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.



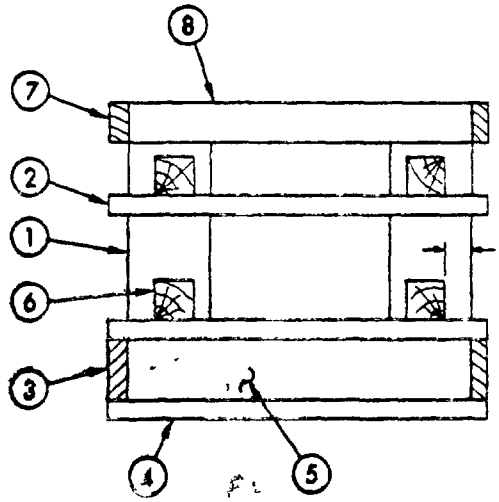
SECTION A-A



DETAIL B
END FRAME
1 REQUIRED



DETAIL D
END FRAME WITH STRUT
REQUIRED

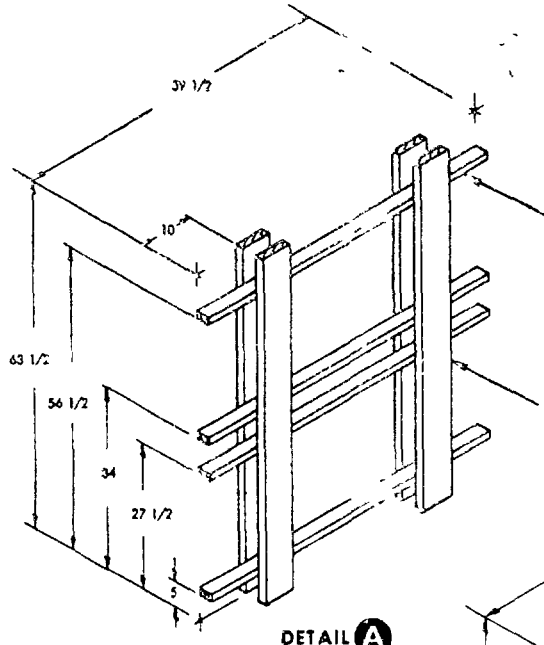


SECTION C-C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

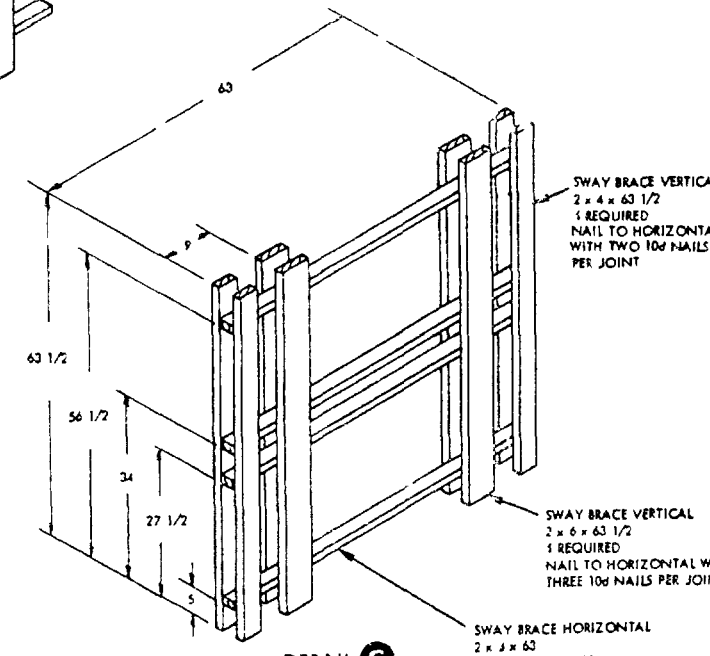
ALTERNATE SWAY BRACE DETAILS

NOTE: THESE ALTERNATE SWAY BRACE DETAILS ARE FOR USE WITH ANY INSIDE WIDTH OF PALLET IF PALLET IS TO BE USED IS OTHER THAN 9'-6" USE HORIZONTAL MEMBER OF APPROPRIATE WIDTH



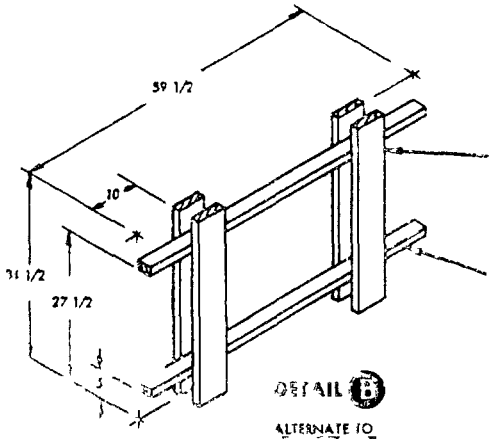
DETAIL A

ALTERNATE TO
DETAIL A - PAGE 6
DETAIL A - PAGE 10 (ADD SPACER WHEN REQUIRED)
DETAIL A - PAGE 14
DETAIL B - PAGE 20
DETAIL A - PAGE 31 (ADD SPACER WHEN REQUIRED)



DETAIL C

ALTERNATE TO
DETAIL C - PAGE 26



DETAIL B

ALTERNATE TO
DETAIL C - PAGE 20
DETAIL A - PAGE 31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

REVIEW ACTIVITY

NAVY OS AS

PREPARING ACTIVITY

NAVY OS

PROJECT NO. 3140-N(00)

INSTRUCTIONS In a continuing effort to make military standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions which may be incorporated in the next revision of the document, or to identify areas needing clarification or interpretation, areas too rigid, restrictive, loose, ambiguous, or which may be incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Ordance Station
Standardization/Documentation Division (Code 5)
Indian Head, MD 20640

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.

(TO BE REMOVED THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER

2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- [ ] VENDOR
- [ ] USER
- [ ] MANUFACTURER
- [ ] OTHER (Specify)

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional

b. WORK TELEPHONE NUMBER (Include Area Code) – Optional

c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional

8. DATE OF SUBMISSION (YYMMDD)

DD FORM 1426

PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:45Z
Check the source to verify that this is the current version before use.